ALUMNI MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

C YASHWANTH

241501251

in partial fulfilment of the End Semester Examination for the course

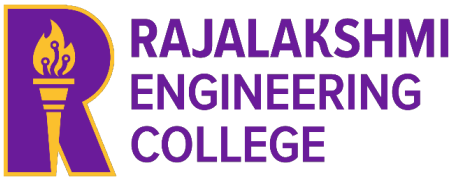
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

Artificial Intelligence and Machine Learning



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

2026-2027

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "Alumni Management System" is the bonafide work of C YASHWANTH (241501251), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE	**SIGNATURE**
Dr. M. Ayyadurai HEAD OF THE DEPARTMENT Department of Artificial Intelligence and Machine Learning Rajalakshmi Engineering College Thandalam – 602 105	Mrs Arulmozhi C COURSE FACULTY Assistant Professor Department of Artificial Intelligence and Machine Learning Rajalakshmi Engineering College

Submitted for the End Semester Examination held on _____.

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, C YASHWANTH (241501251), declare that this report titled “Alumni Management System” documents the application in my Zoho Creator account for the course CB23F35 – Low-Code Application Development using Zoho Creator. The report identifies the application components, available evidence and remaining verification work.

The application was built and deployed on the Zoho Creator platform with AI-assisted development. AI assistance was also used to prepare this report. These contributions and the platform documentation are acknowledged. I am responsible for reviewing the report, understanding the implementation and demonstrating the application. This declaration is to be signed only after my review.

Place: Chennai
Date: _____

SIGNATURE OF THE STUDENT
C YASHWANTH
241501251

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman, Mr Meganathan M, Chairperson, Ms Thangam Meganathan and Vice-Chairman, Mr Abhay Meganathan for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr Murugesan SN, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. M. Ayyadurai, Head of the Department of Artificial Intelligence and Machine Learning, for the guidance and support provided throughout the course.

I sincerely thank Mrs Arulmozhi C, Assistant Professor and course faculty, Department of Artificial Intelligence and Machine Learning, for her guidance, reviews and feedback during the development of this application and report.

I also acknowledge Zoho Corporation / Zoho Creator Education Programme for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

ABSTRACT

Alumni relations offices maintain graduate contact details, organise reunions and webinars, raise funds for scholarships and connect current students with experienced alumni. When alumni details, event registrations, donations, job referrals and mentorship requests are recorded in separate spreadsheets, forms and messaging groups, staff find it difficult to keep records verified, avoid duplicate entries and follow up on pending work. The Alumni Management System centralizes these details in a Zoho Creator application.

The Alumni Management System was built in Zoho Creator. The application includes Departments, Alumni, Events, Event Registrations, Fundraising Campaigns, Donations, Job Postings, Mentorship Requests and Success Stories forms, a Dashboard page, list, calendar and Kanban reports, a mentorship_lifecycle blueprint, Deluge workflows, user roles and an email notification that informs the alumni office when a new alumnus registers.

The screenshots document the Alumni and related forms, validation messages, the registration email notification, the live Dashboard with counters and charts, reports, the blueprint, the user permission profiles and the deployment of version 1.0 to the Stage and Production environments.

Keywords: Low-Code Development; Zoho Creator; Alumni Management; Deluge; Workflow Automation; Mentorship Lifecycle

PROJECT AT A GLANCE

Attribute	Details
Project Title	Alumni Management System
Application Domain	Education – Alumni Relations
Student Name & Register No.	C YASHWANTH – 241501251
Zoho Creator Edition	Trial account
Application Name (Link Name)	Alumni Management System (alumni-management-system)
Published / Live URL	https://creatorapp.zoho.in/yashwanth.c.2024.aiml_raj/alumni-management-system/
Demo Credentials (per role)	Not included; use assigned Creator accounts
Number of Forms / Subforms	9 forms / 1 subform (Employment History in Alumni)
Number of Reports (by type)	List, calendar and Kanban reports for all nine forms
Number of Pages / Dashboards	1 dashboard page evidenced
Number of Workflows	3 form workflows (DS-01 to DS-03); 1 blueprint
Deluge Functions / Custom Actions	Registration email, donation validation and donation notification scripts; no standalone function evidenced
Integrations	Email through Deluge sendmail; Zoho Cliq and Zoho Calendar connections configured
Roles & Profiles Defined	Administrator; Alumni Coordinator and Finance Officer profiles; Read and Write profiles; portal permissions for alumni and students
Environments Used	Development, Stage and Production (version 1.0)
Date of Production Deployment	21 October 2026

TABLE OF CONTENTS

ABSTRACT....iv
PROJECT AT A GLANCE....v
LIST OF TABLES....viii
LIST OF FIGURES....ix
LIST OF ABBREVIATIONS....xi

CHAPTER 1 – INTRODUCTION....1
1.1 Background....1
1.2 Problem Statement....1
1.3 Objectives....1
1.4 Scope of the Project....2
1.5 Justification for a Low-Code Approach....2
1.6 Organisation of the Report....2

CHAPTER 2 – LITERATURE SURVEY AND SYSTEM STUDY....3
2.1 Study of the Existing System....3
2.2 Limitations of the Existing System....3
2.3 Low-Code versus Traditional Development....3
2.4 Literature Survey....4
2.5 Emerging Trends in Low-Code Development....5
2.6 Proposed System....5

CHAPTER 3 – SYSTEM ANALYSIS AND REQUIREMENTS....7
3.1 Stakeholders and User Roles....7
3.2 Functional Requirements....7
3.3 Non-Functional Requirements....8
3.4 Hardware and Software Requirements....8
3.5 Use Case Model....8

CHAPTER 4 – SYSTEM DESIGN....10
4.1 System Architecture....10
4.2 Data Model....10
4.3 Form Design....12
4.4 Process and Workflow Design....18
4.5 User Interface and Navigation Design....19
4.6 Access Control Design....20

CHAPTER 5 – IMPLEMENTATION....22
5.1 Application Setup....22
5.2 Forms and Data Capture....23
5.3 Reports....28
5.4 Pages and Dashboards....31
5.5 Deluge Scripting....32
5.6 Workflows and Automation....33
5.7 Integrations....35
5.8 User Management and Role-Based Access....35

5.9 AI and Analytics Features....36
5.10 Pro-Code Extension....36
5.11 Mobile and Portal Access....36
CHAPTER 6 – TESTING AND VALIDATION....37
6.1 Testing Strategy....37
6.2 Test Cases....37
6.3 Debugging Deluge Scripts....38
6.4 Defect Log....38
6.5 User Acceptance Testing....38
CHAPTER 7 – DEPLOYMENT, SECURITY AND GOVERNANCE....39
7.1 Environments and Deployment Strategy....39
7.2 Performance Optimisation....41
7.3 Security Features....41
7.4 Governance and Administration....41
CHAPTER 8 – RESULTS AND DISCUSSION....42
8.1 Application Output....42
8.2 Achievement of Objectives....43
8.3 Comparison with the Existing System....43
8.4 Limitations of the Low-Code Approach....44
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum....44
CHAPTER 9 – CONCLUSION AND FUTURE ENHANCEMENTS....45
9.1 Conclusion....45
9.2 Future Enhancements....45
9.3 Learning Outcomes and Career Relevance....45
REFERENCES....46
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST....47
APPENDIX B – COURSE OUTCOME MAPPING....49
APPENDIX C – COMPLETE DELUGE SOURCE CODE....50
C.1 Alumni Workflow – Registration Email Notification....50
C.2 Donations Workflow – Amount and Date Validation....50
C.3 Donations Workflow – Donation Notification Email....50
APPENDIX D – USER MANUAL....51
D.1 Accessing the Application....51
D.2 Instructions for Administrator and Alumni Coordinator....51
D.3 Instructions for Finance Officer, Alumni and Students....51
APPENDIX E – EVALUATION RUBRIC....52
APPENDIX F – VIVA-VOCE RECORD SHEET....54

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application 4
Table 2.2: Summary of Literature Survey 5
Table 3.1: Stakeholders and User Roles 7
Table 3.2: Functional Requirements 7
Table 3.3: Non-Functional Requirements 8
Table 3.4: Hardware and Software Requirements 8
Table 3.5: Use Case Description 9
Table 4.1: Entities and Relationships 11
Table 4.2: Field Specification for Departments 13
Table 4.3: Field Specification for Alumni 13
Table 4.4: Field Specification for Events 14
Table 4.5: Field Specification for Event Registrations 15
Table 4.6: Field Specification for Fundraising Campaigns 16
Table 4.7: Field Specification for Donations 16
Table 4.8: Field Specification for Job Postings 17
Table 4.9: Field Specification for Mentorship Requests 17
Table 4.10: Field Specification for Success Stories 18
Table 4.11: Role–Permission Matrix 21
Table 5.1: Reports Implemented in the Application 28
Table 5.2: Deluge Scripts Implemented 33
Table 5.3: Workflows and Blueprint Processes 34
Table 5.4: Integrations Implemented 35
Table 6.1: Functional and Workflow Test Cases 37
Table 6.2: Defect Log 38
Table 6.3: User Acceptance Feedback 38
Table 7.1: Deployment Checklist 40
Table 7.2: Performance Optimisation Results 41
Table 7.3: Security Controls Implemented 41
Table 8.1: Mapping of Objectives to Outcomes 43
Table 8.2: Existing System versus Proposed Application 43
Table A.1: Feature Coverage Checklist 47
Table B.1: Mapping of Course Outcomes to Report Chapters 49
Table E.1: Evaluation Rubric for the Individual Project 52
Table E.2: Performance Level Descriptors 53
Table F.1: Viva-Voce Questions and Responses 54
Table F.2: Marks Summary 54

LIST OF FIGURES

Figure 2.1: Alumni Office Process Used as the Baseline 3
Figure 3.1: Alumni Management System Use Cases and Actors 9
Figure 4.1: Alumni Management System Architecture 10
Figure 4.2: Department, Alumni and Activity Relationships 12
Figure 4.3: Alumni Registration and Notification Workflow 19
Figure 4.4: Alumni registration email notification received in the mailbox 19
Figure 4.5: Application Navigation Structure 20
Figure 4.6: Dashboard page in the Zoho Creator page builder 20
Figure 5.1: Alumni Management System in the Zoho Creator Applications workspace 22
Figure 5.2: Live application home with the Dashboard and navigation menu 22
Figure 5.3: Alumni form in live mode – personal details 23
Figure 5.4: Alumni form – academic details with department lookup 23
Figure 5.5: Alumni form – engagement and office-use fields 24
Figure 5.6: Events form with schedule, venue and type details 24
Figure 5.7: Events form – capacity, registration deadline, fee and status 25
Figure 5.8: Donations form with alumni and campaign lookups 25
Figure 5.9: Donations form – status, 80G receipt and PAN fields 25
Figure 5.10: Mentorship Requests form with student and mentor details 26
Figure 5.11: Mentorship Requests form with the Sessions subform 26
Figure 5.12: Alumni form in the builder with field properties 27
Figure 5.13: Alumni form builder – academic and professional fields 27
Figure 5.14: Donations form validation message on submission 27
Figure 5.15: Alumni Report in live mode 29
Figure 5.16: Alumni grouped by industry in a Kanban report 29
Figure 5.17: Donations grouped by payment mode in a Kanban report 29
Figure 5.18: Events calendar report for October 2026 30
Figure 5.19: Fundraising Campaigns report with goal, amount raised and progress 30
Figure 5.20: Job Postings report with job type and required skills 30
Figure 5.21: Success Stories report with publish status 31
Figure 5.22: Dashboard page in the Zoho Creator page builder 31
Figure 5.23: Live Dashboard with counters, charts and quick links 32
Figure 5.24: Dashboard with the Donation Status chart and embedded Event Registrations report 32
Figure 5.25: Alumni registration email notification received in Gmail 34
Figure 5.26: Blueprints page showing the enabled mentorship_lifecycle blueprint 34
Figure 5.27: mentorship_lifecycle blueprint with stages and transitions 35
Figure 5.28: User Permissions page with the Alumni Coordinator and Finance Officer profiles 36
Figure 6.1: Empty-submit validation errors in the Alumni form 38
Figure 7.1: Environments page with version 1.0 published to Stage and Production. 39
Figure 7.2: Publish menu with the Publish to Stage and Publish to Production options . 39

Figure 7.3: Development environment of the application with no records 40
Figure 7.4: Authenticated live application with the signed-in user and application URL 40
Figure 8.1: Alumni form in live mode 42
Figure 8.2: Alumni registration email notification 42
Figure 8.3: Live Dashboard with donation and campaign charts 43

LIST OF ABBREVIATIONS

The following abbreviations are used in this report.

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
INR	Indian Rupee
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
PAN	Permanent Account Number
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
Zoho Creator	Low-code application platform

CHAPTER 1: INTRODUCTION

1.1 Background

The project addresses the day-to-day work of a college alumni relations office. Alumni are registered through the Alumni form and linked to their department, events are planned and registrations recorded, fundraising campaigns collect donations, alumni share job openings, and current students request mentorship from experienced graduates. Status-based reports and a dashboard help staff review pending verifications, open events, donations and mentorship requests.

1.2 Problem Statement

The project addresses fragmented alumni and engagement information. Without a common application, staff may need to search separate spreadsheets and message threads to confirm an alumnus's contact details, check whether an event is full, total the donations to a campaign or find which mentorship requests are still waiting for a response.

- Alumni, department, event, donation, job and mentorship details are stored in separate records and must be related consistently.
- Verification, approval and mentorship progress are represented through status fields and process stages, so users need to consult the corresponding report or board to understand the current state.
- Notifications to the alumni office depend on correctly entered alumni details and on workflows that run when a record is submitted.

1.3 Objectives

- To create structured forms for departments, alumni, events, event registrations, fundraising campaigns, donations, job postings, mentorship requests and success stories.
- To organise verification, event, donation, job and mentorship progress through status fields and provide clear visibility of pending work.
- To provide reports and a Dashboard page for alumni registration, event and donation monitoring and mentorship tracking.
- To relate records using department, alumni, event and campaign lookups and to automate validation and email notification through Deluge.
- To document testing, access control and deployment requirements and distinguish verified configuration from work that remains unverified.

1.4 Scope of the Project

In scope are the nine forms, their related lookups, the list, calendar and Kanban reports, the Dashboard page, the alumni registration email workflow, the donation validation and notification workflows, the mentorship_lifecycle blueprint, the permission profiles and the deployment of version 1.0 to Stage and Production. The screenshots show a web application interface. Mobile-specific behaviour, customer portal access for alumni and students and payment gateway collection are outside the verified evidence for this report.

1.5 Justification for a Low-Code Approach

Zoho Creator suits this application because its visual form and report builders support the linked records and status views required by an alumni office, while Deluge extends event-driven behaviour for validation and email notifications. A blueprint provides a visual way to represent the ordered stages of a mentorship. This combination lets the project demonstrate data capture, reporting and workflow automation without building a separate web stack. The platform's managed hosting and mobile capabilities are platform features; the supplied evidence does not establish that mobile access was configured in this application [1], [2].

1.6 Organisation of the Report

Chapter 2 studies the manual baseline, low-code development and related sources. Chapter 3 records project requirements. Chapter 4 presents the data model, processes and access design. Chapter 5 describes the Zoho Creator implementation. Chapter 6 defines the verification evidence and test plan. Chapter 7 documents deployment, security and governance status. Chapter 8 reviews the demonstrated outputs and limitations. Chapter 9 concludes the report and lists future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The baseline used for comparison is a typical manual process in a college alumni office; no separate legacy application or measured baseline was supplied. An alumnus submits details through an online form or email. Office staff copy the details into a spreadsheet, announce events in messaging groups, collect registrations through a separate form, note donations in a ledger and forward job openings informally. Mentorship requests reach alumni through personal contacts. This process description is a project assumption used for analysis, not an audited description of a named organisation.

Alumnus submits details → office staff copy them into a spreadsheet → events announced and registrations collected separately → donations noted in a ledger → job openings and mentorship requests forwarded informally → reports compiled by hand. Each handoff depends on a person updating a separate record.

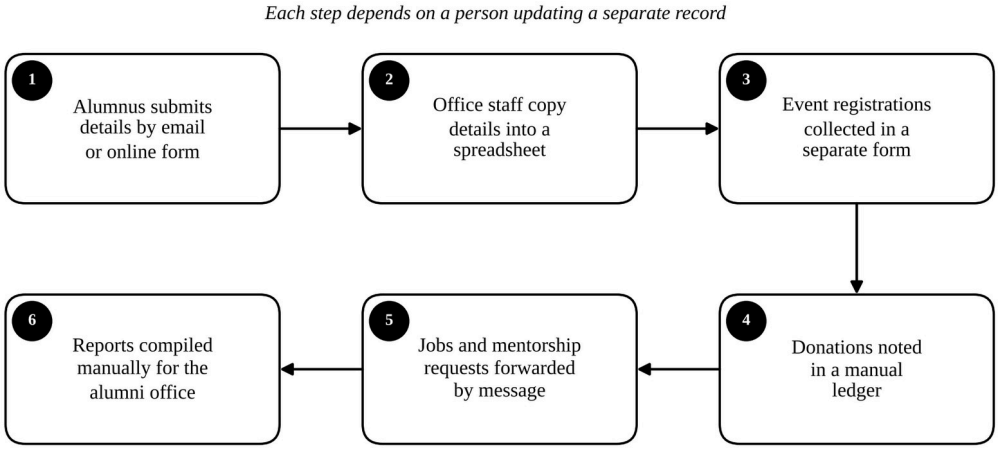


Figure 2.1: Alumni Office Process Used as the Baseline

2.2 Limitations of the Existing System

- Alumni records may be duplicated or become inconsistent across separate files, and no record shows whether an alumnus has been verified.
- Event capacity, campaign totals and mentor availability depend on manual counting, which can make outstanding work less visible.
- Approvals for job postings and the progress of mentorship requests depend on manual follow-up in messages.

2.3 Low-Code versus Traditional Development

For this project, Zoho Creator low-code development provides reusable form, lookup, report and workflow configuration in one application. A traditional application would

require separate implementation of its database, user interface, hosting, authentication and deployment pipeline. The low-code route reduces the amount of infrastructure code but still requires careful data modelling, Deluge scripting, permission review and platform-specific configuration. No comparative development-time measurement was recorded; therefore the comparison is qualitative.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Separate implementation of data, UI, hosting and workflows; no estimate measured.	Visual form, report and workflow configuration reduces custom infrastructure code; no estimate measured.
Skills required	Programming, database, web and deployment skills.	Creator configuration, data modelling and Deluge skills.
Hosting & infrastructure	Selected and maintained by development team.	Provided as part of Creator; app-specific setup still required.
Mobile application	Requires responsive UI or separate client work.	Creator offers mobile access; this project tested web screens only.
Security & access control	Implemented through selected identity and authorization components.	Platform supports roles and permissions; separate-role tests pending.
Maintenance & change	Changes require code, build and deployment practices.	Forms and reports can be changed visually; Deluge and platform dependencies remain.
Customisation limits	Broad control over architecture and UI.	Faster configuration within platform capabilities and limits.
Cost	Depends on engineering, hosting and maintenance choices.	Depends on subscription and configuration effort; actual project cost not recorded.

2.4 Literature Survey

The survey below uses two peer-reviewed review papers and three primary Zoho Creator documentation sources. The literature supports a balanced assessment: low-code can accelerate delivery, while complexity, governance and platform constraints remain design considerations [1], [2]. Zoho's documentation describes Creator workflows and blueprints as configurable mechanisms for event handling and staged processes [3]–[5].

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Gao et al., 2026	Systematic review of low-code viability	Reports development-speed benefits with context-dependent quality and complexity outcomes.	Supports balanced evaluation; does not prove this project's performance.
2	Ajimati et al., 2025	Adoption of low-code and no-code development	Reviews adoption drivers and governance concerns.	Frames platform choice and governance for the alumni office.
3	Zoho Corporation	Zoho Creator Quickstart	Describes forms, reports, pages and workflows.	Basis for the nine forms and the Dashboard page.
4	Zoho Corporation	Understanding workflows	Explains form events and workflow actions.	Registration email and donation validation workflows.
5	Zoho Corporation	Creating blueprints	Describes stages and transitions.	Mentorship Lifecycle blueprint.

2.5 Emerging Trends in Low-Code Development

Low-code platforms increasingly combine visual builders with scripting, analytics and AI-assisted features. In Zoho Creator, form and report builders handle most structure, Deluge handles rules that exceed field configuration, and built-in AI assistance can suggest fields and generate components that the developer then reviews. Citizen-developer maintainability depends on clear field names and reports, while rules such as duplicate checks and notifications remain explicit Deluge code that can be read and tested.

2.6 Proposed System

The proposed system stores departments, alumni profiles, events, event registrations, fundraising campaigns, donations, job postings, mentorship requests and success stories in nine related Zoho Creator forms. Lookup fields connect every activity to the alumnus who performed it, so a single record shows an alumnus's registrations, donations, postings and mentoring.

- Verification status and office-use fields give the alumni office one verified register of graduates.
- List, calendar and Kanban reports organise daily work by status, date and category.

- A Dashboard page combines counters, charts, quick links and an embedded registrations report.
- Deluge workflows validate entries and send email notifications automatically, and a blueprint models staged processes.
- Permission profiles separate the responsibilities of the coordinator, finance officer, alumni and students.

CHAPTER 3
SYSTEM ANALYSIS AND REQUIREMENTS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Responsibilities
Administrator	Application owner (Alumni Relations Head)	Manage all modules, users, roles and settings
Alumni Coordinator	Alumni office staff	Verify alumni, manage events, review job postings and success stories, monitor mentorships
Finance Officer	Fund management	Manage campaigns and donations; review donation reports
Alumni	Graduate of the college	Maintain profile, register for events, donate, post jobs, mentor students
Student	Current student	View mentors and open jobs; submit mentorship requests

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Roles	Priority
FR-01	Register alumni with unique email, phone and register number	Administrator, Coordinator, Alumni	High
FR-02	Verify or reject alumni profiles	Administrator, Coordinator	High
FR-03	Maintain department master data used in lookups	Administrator	High
FR-04	Create events with capacity, deadline, mode and venue	Administrator, Coordinator	High
FR-05	Register alumni for events and record attendance	Alumni, Coordinator	High
FR-06	Create fundraising campaigns with goal amount and dates	Finance Officer	High
FR-07	Record donations against campaigns with validation	Finance Officer, Alumni	High
FR-08	Post job openings with review status	Alumni, Coordinator	Medium
FR-09	Request and track mentorship through defined stages	Student, Alumni	High
FR-10	Submit and publish success stories	Alumni, Coordinator	Medium
FR-11	Provide list, Kanban and summary reports	Permitted roles	High
FR-12	Provide a Dashboard page with counters and charts	Administrator, Coordinator	High

FR ID	Requirement	Roles	Priority
FR-13	Send email notifications on registration and donation	System	Medium
FR-14	Restrict module actions by role	All roles	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement and Verification Status
NFR-01	Performance	Operational reports should remain usable at expected volume; benchmark not measured
NFR-02	Security	Users must access only permitted modules and records; separate-role tests pending
NFR-03	Usability	Consistent navigation and clear required fields; browser observed
NFR-04	Availability	Authenticated hosted access required; inspection session verified
NFR-05	Maintainability	Consistent link names and readable Deluge scripts
NFR-06	Privacy	Alumni contact details visible only to permitted roles

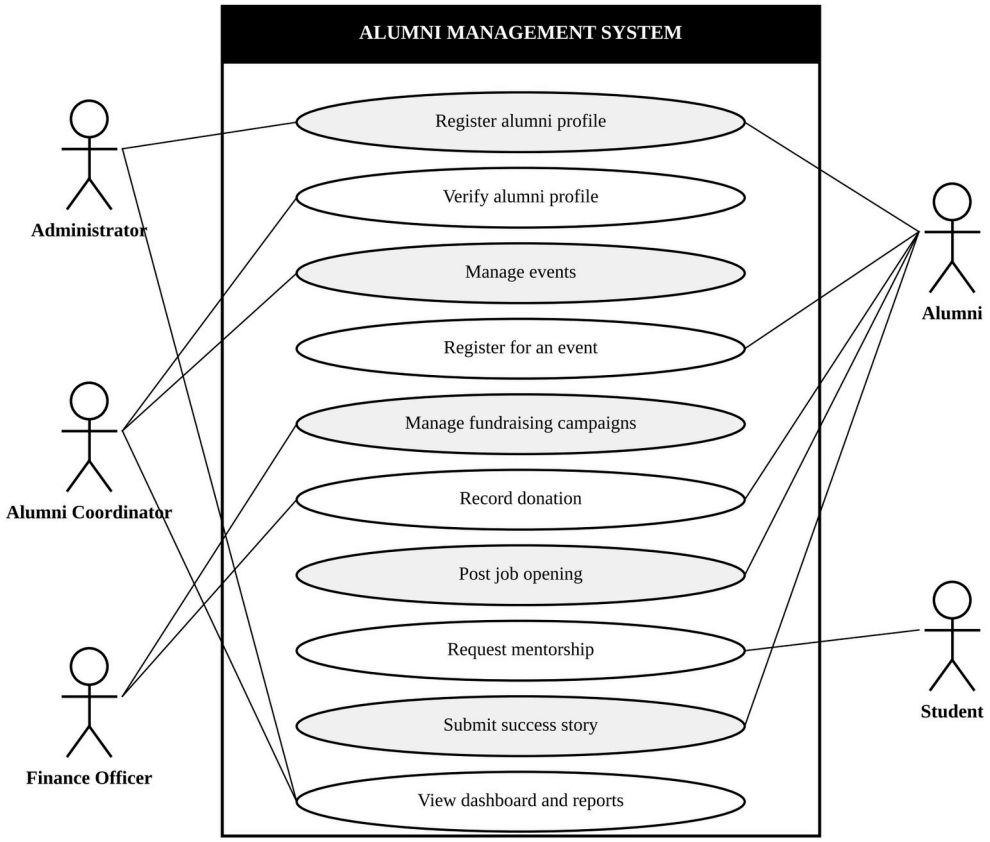
3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (India data centre), trial account
Application scripts	Deluge
Browser	Google Chrome on Windows
Email	Deluge sendmail to the administrator mailbox
Other services	Zoho Cliq and Zoho Calendar connections configured
Mobile	Native mobile application and portal not verified
Client environment	Windows laptop and internet connection; no minimum hardware benchmark

3.5 Use Case Model

The main actors are the Administrator, Alumni Coordinator, Finance Officer, Alumni and Student. The use case below describes alumni registration, which triggers the email notification demonstrated in this report.



Field	Description
Use Case ID	UC-01
Use Case Name	Register an alumnus
Actors	Alumni, Alumni Coordinator
Precondition	User can open the Alumni form; department records exist
Main Flow	Enter personal, academic and professional details; select department; submit; record saved with Pending Verification status; email sent to the alumni office
Alternative Flow	Missing mandatory fields or invalid values are rejected with validation messages
Postcondition	Alumni record stored and visible in alumni reports; notification email received
Related Requirements	FR-01, FR-02, FR-13

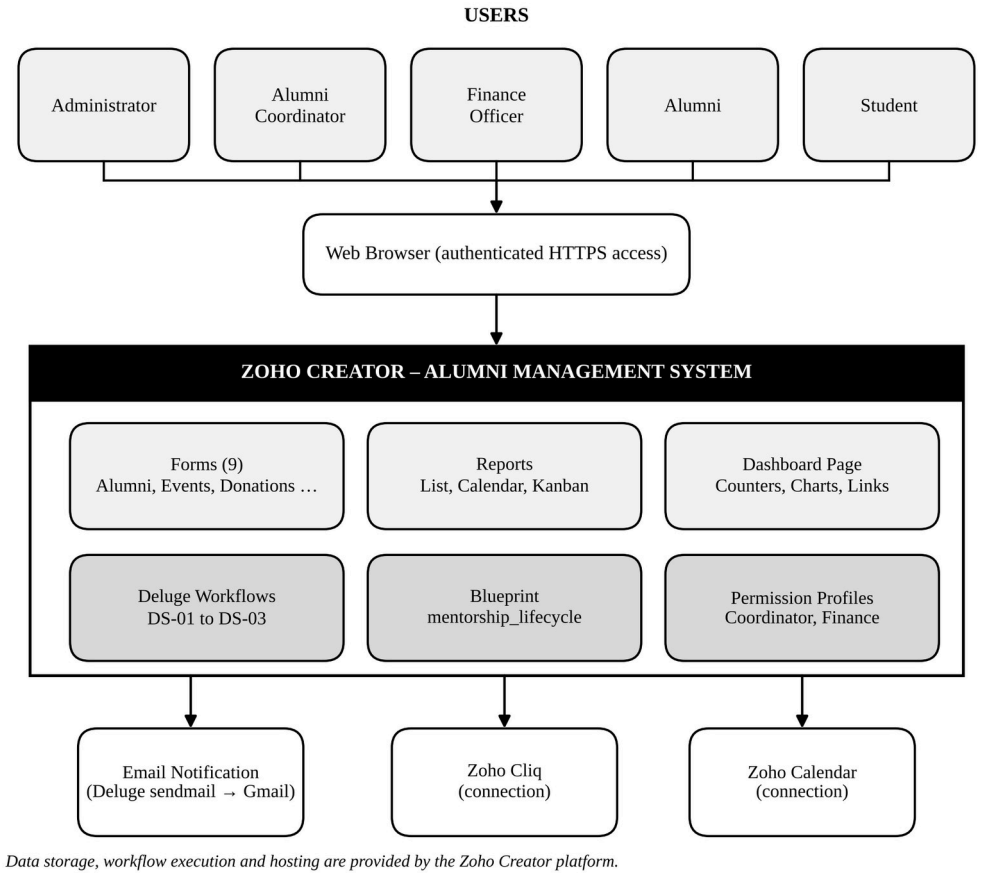
Figure 3.1: Alumni Management System Use Cases and Actors

Table 3.5: Use Case Description

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The browser presents the forms, reports and Dashboard page to each role. Zoho Creator stores the data and executes the Deluge workflows and the blueprint. Email notifications are sent through Deluge sendmail to the alumni office mailbox, and Zoho Cliq and Zoho Calendar connections are configured for team messaging and event dates.



Data storage, workflow execution and hosting are provided by the Zoho Creator platform.

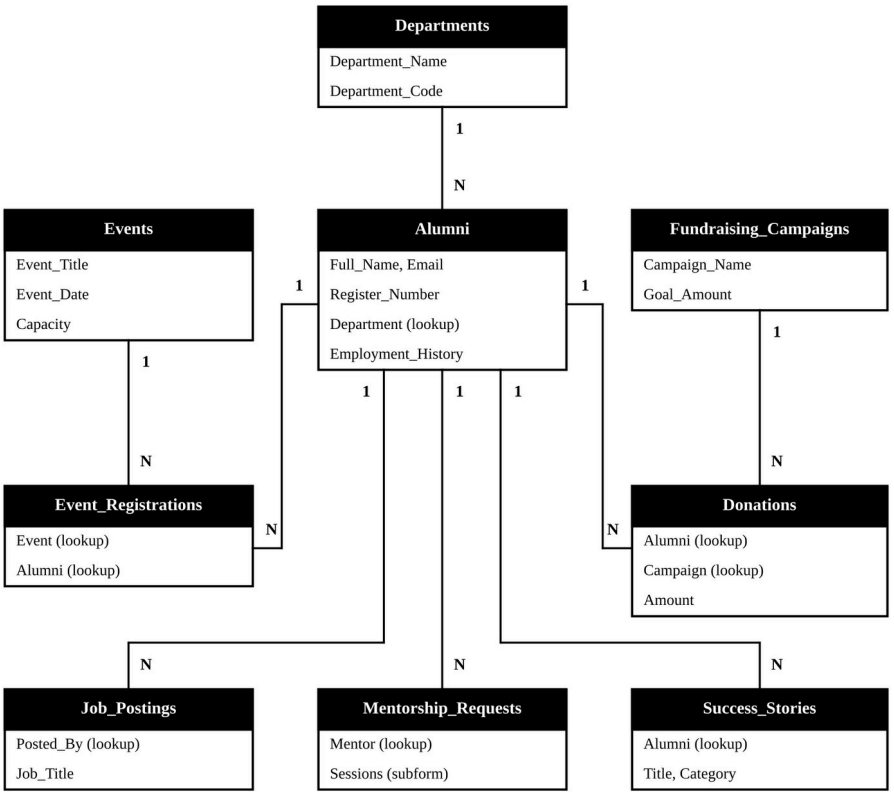
Figure 4.1: Alumni Management System Architecture

4.2 Data Model

The data model uses Alumni as the central entity. Departments are referenced by alumni, events and mentorship requests, and alumni are referenced by event registrations, donations, job postings, mentorship requests and success stories.

Table 4.1: Entities and Relationships

Entity	Key Fields	Related Entity	Relationship
Departments	Department_Name, Department_Code	Alumni, Events, Mentorship_Requests	One to many
Alumni	Full_Name, Email, Register_Number	Event_Registrations, Donations, Job_Postings, Mentorship_Requests, Success_Stories	One to many
Events	Event_Title, Event_Date	Event_Registrations	One to many
Event_Registrations	Event, Alumni	Events and Alumni	Many to one (two lookups)
Fundraising_Campaigns	Campaign_Name, Goal_Amount	Donations	One to many
Donations	Amount, Donation_Date	Campaigns and Alumni	Many to one (two lookups)
Job_Postings	Company_Name, Job_Title	Alumni	Many to one
Mentorship_Requests	Student_Name, Mentor	Alumni, Departments	Many to one
Success_Stories	Title, Category	Alumni	Many to one



1 – N : one record relates to many records through a lookup field

Figure 4.2: Department, Alumni and Activity Relationships

4.3 Form Design

Each form was designed with clear field names, mandatory markers and link names that Deluge scripts use. The Alumni form groups its fields into personal, academic, professional, engagement and office-use sections, and the related forms reuse the department, alumni, event and campaign lookups so that names are never typed twice. Tables 4.2 to 4.10 give the field specification of every form; the screenshots in Chapter 5 show the forms in live mode.

Table 4.2: Field Specification for Departments

Field	Link Name	Type	Required	Rule / Choices
Department Name	Department_Name	Single Line	Yes	Unique
Department Code	Department_Code	Single Line	Yes	Unique, e.g. AIML, CSE
HOD Name	HOD_Name	Single Line	No	—
HOD Email	HOD_Email	Email	No	—
Alumni Coordinator Email	Alumni_Coordinator_Email	Email	No	Receives department notifications
Is Active	Is_Active	Decision Box	No	Ticked by default

Table 4.3: Field Specification for Alumni

Field	Link Name	Type	Required	Rule / Choices
Alumni ID	Alumni_ID	Single Line	No	No duplicate values
Full Name	Full_Name	Single Line	Yes	—
Gender	Gender	Radio	Yes	Male, Female, Other
Date of Birth	Date_of_Birth	Date	Yes	Past date
Email	Email	Email	Yes	Unique
Phone	Phone	Phone	Yes	Encrypted field
LinkedIn Profile	LinkedIn_Profile	URL	No	—
Profile Photo	Profile_Photo	Image	No	—
Register Number	Register_Number	Single Line	Yes	Unique
Department	Department	Lookup	Yes	Departments
Programme	Programme	Drop Down	Yes	B.E., B.Tech, M.E., M.Tech, MBA, MCA, Ph.D
Batch Start Year	Batch_Start_Year	Number	Yes	—
Graduation Year	Graduation_Year	Number	Yes	After batch start year
CGPA	CGPA	Decimal	No	0 to 10

Degree Certificate	Degree_Certificate	Image	No	—

Field	Link Name	Type	Required	Rule / Choices
Employment Status	Employment_Status	Drop Down	Yes	Employed, Higher Studies, Entrepreneur, Government Service, Seeking Opportunities, Other
Current Organisation	Current_Organisation	Single Line	No	—
Designation	Designation	Single Line	No	—
Industry	Industry	Drop Down	No	IT Services, Banking and Finance, Consulting, Healthcare, Government and PSU and others
Employment History	Employment_History	Subform	No	Organisation, role and dates
Willing to Mentor	Willing_to_Mentor	Decision Box	No	—
Max Mentees	Max_Mentees	Number	No	1 to 5
Membership Type	Membership_Type	Drop Down	Yes	Basic, Lifetime Member
Verification Status	Verification_Status	Drop Down	No	Pending Verification, Verified, Rejected
Profile Completeness	Profile_Completeness	Percent	No	Office use
Total Donated	Total_Donated	Currency	No	Office use

Table 4.4: Field Specification for Events

Field	Link Name	Type	Required	Rule / Choices
Event Title	Event_Title	Single Line	Yes	—

Event Type	Event_Type	Drop Down	Yes	Alumni Reunion, Webinar, Guest Lecture, Workshop, Networking Meet, Career Fair, Sports Meet
Event Mode	Event_Mode	Radio	Yes	Hybrid, Offline, Online
Event Date	Event_Date	Date	Yes	—
Start Time	Start_Time	Time	No	—

Field	Link Name	Type	Required	Rule / Choices
End Date	End_Date	Date	Yes	Not before event date
Venue	Venue	Single Line	No	For offline and hybrid events
Target Department	Target_Department	Lookup	No	Departments
Capacity	Capacity	Number	Yes	—
Registered Count	Registered_Count	Number	No	Seats booked
Seats Available	Seats_Available	Number	No	Capacity less registered
Registration Deadline	Registration_Deadline	Date	Yes	Not after event date
Registration Fee	Registration_Fee	Currency	No	Per person
Event Status	Event_Status	Drop Down	Yes	Planned, Open for Registration, Registration Closed, Completed, Cancelled
Organiser Email	Organiser_Email	Email	No	—
Event Banner	Event_Banner	File Upload	No	—

Table 4.5: Field Specification for Event Registrations

Field	Link Name	Type	Required	Rule / Choices
Registration No	Registration_No	Auto Number	No	System generated
Event	Event	Lookup	Yes	Events
Alumni	Alumni	Lookup	Yes	Alumni
Registration Date	Registration_Date	Date	Yes	—

Number of Guests	Number_of_Guests	Number	No	—
Fee Payable	Fee_Payable	Currency	No	—
Payment Status	Payment_Status	Drop Down	Yes	Not Applicable, Pending, Paid
Attendance Status	Attendance_Status	Drop Down	Yes	Registered, Attended, No Show, Cancelled
Feedback Rating	Feedback_Rating	Number	No	1 to 5
Feedback Comments	Feedback_Comments	Multi Line	No	—

Table 4.6: Field Specification for Fundraising Campaigns

Field	Link Name	Type	Required	Rule / Choices
Campaign No	Campaign_No	Auto Number	No	System generated
Campaign Name	Campaign_Name	Single Line	Yes	Unique
Purpose	Purpose	Drop Down	Yes	Scholarship Fund, Infrastructure, Research Lab, Student Welfare, Sports, Library, General Fund
Goal Amount	Goal_Amount	Currency	Yes	—
Amount Raised	Amount_Raised	Currency	No	—
Progress (%)	Progress_Percent	Percent	No	—
Donor Count	Donor_Count	Number	No	—
Start Date	Start_Date	Date	Yes	—
End Date	End_Date	Date	Yes	After start date
Campaign Status	Campaign_Status	Drop Down	Yes	Draft, Active, Goal Achieved, Closed

Table 4.7: Field Specification for Donations

Field	Link Name	Type	Required	Rule / Choices
Alumni	Alumni	Lookup	Yes	Alumni
Campaign	Campaign	Lookup	Yes	Fundraising_Campaigns
Amount	Amount	Currency	Yes	Minimum 100 (DS-02)
Donation Date	Donation_Date	Date	Yes	Not in the future (DS-02)
Payment Mode	Payment_Mode	Drop Down	Yes	UPI, Net Banking, Card, Cheque, Cash
Transaction Reference	Transaction_Reference	Single Line	No	—
Donation Status	Donation_Status	Drop Down	Yes	Pledged, Received, Cancelled
Need 80G Receipt	Need_80G_Receipt	Decision Box	No	—
PAN Number	PAN_Number	Single Line	No	For 80G receipts
Anonymous Donation	Anonymous_Donation	Decision Box	No	—
Receipt Sent	Receipt_Sent	Decision Box	No	—

Remarks	Remarks	Multi Line	No	—

Table 4.8: Field Specification for Job Postings

Field	Link Name	Type	Required	Rule / Choices
Job No	Job_No	Auto Number	No	System generated
Posted By	Posted_By	Lookup	Yes	Alumni
Company Name	Company_Name	Single Line	Yes	—
Job Title	Job_Title	Single Line	Yes	—
Job Type	Job_Type	Drop Down	Yes	Full-time, Internship, Part-time, Contract, Freelance
Work Mode	Work_Mode	Radio	Yes	On-site, Remote, Hybrid
Experience Required	Experience_Required	Drop Down	Yes	Fresher, 0-2 Years, 2-5 Years, 5+ Years
Skills Required	Skills_Required	Multi Select	No	Python, Java, Data Science, Cloud Computing and others
Application Deadline	Application_Deadline	Date	Yes	—
Approval Status	Approval_Status	Drop Down	Yes	Pending Review, Approved, Rejected, Expired, Filled

Table 4.9: Field Specification for Mentorship Requests

Field	Link Name	Type	Required	Rule / Choices
Student Name	Student_Name	Single Line	Yes	—
Student Register Number	Student_Register_Number	Single Line	Yes	—
Student Email	Student_Email	Email	Yes	—
Student Department	Student_Department	Lookup	Yes	Departments
Student Year	Student_Year	Drop Down	Yes	I Year to IV Year
Mentor	Mentor	Lookup	Yes	Alumni

Mentorship Area	Mentorship_Area	Drop Down	Yes	Career Guidance, Higher Studies, Interview Preparation, Entrepreneurship, Technical Skills, Research

Field	Link Name	Type	Required	Rule / Choices
Preferred Mode	Preferred_Mode	Radio	Yes	Phone, Online, In-person
Message to Mentor	Message	Multi Line	Yes	—
Request Date	Request_Date	Date	Yes	—
Request Status	Request_Status	Drop Down	Yes	Requested, Accepted, Declined, In Progress, Completed
Sessions	Sessions	Subform	No	One row per mentoring session
Total Sessions	Total_Sessions	Number	No	—
Student Rating	Student_Rating	Number	No	1 to 5

Table 4.10: Field Specification for Success Stories

Field	Link Name	Type	Required	Rule / Choices
Story No	Story_No	Auto Number	No	System generated
Alumni	Alumni	Lookup	Yes	Alumni
Title	Title	Single Line	Yes	—
Category	Category	Drop Down	Yes	Award or Recognition, Promotion, Startup Launch, Publication or Patent, Higher Studies Admission, Social Impact
Achievement Date	Achievement_Date	Date	Yes	Not in the future
Story	Story	Multi Line	Yes	—
Photo	Photo	Image	No	—

Publish Status	Publish_Status	Drop Down	Yes	Submitted, Published, Rejected

4.4 Process and Workflow Design

When an alumnus submits the Alumni form, the record is saved with Pending Verification status and an on-success workflow emails the alumni office. Donations are validated before saving and a notification is sent once recorded. Mentorship requests move through the Request_Status values Requested, Accepted or Declined, In Progress and Completed, and a blueprint named mentorship_lifecycle defines staged transitions for alumni records.

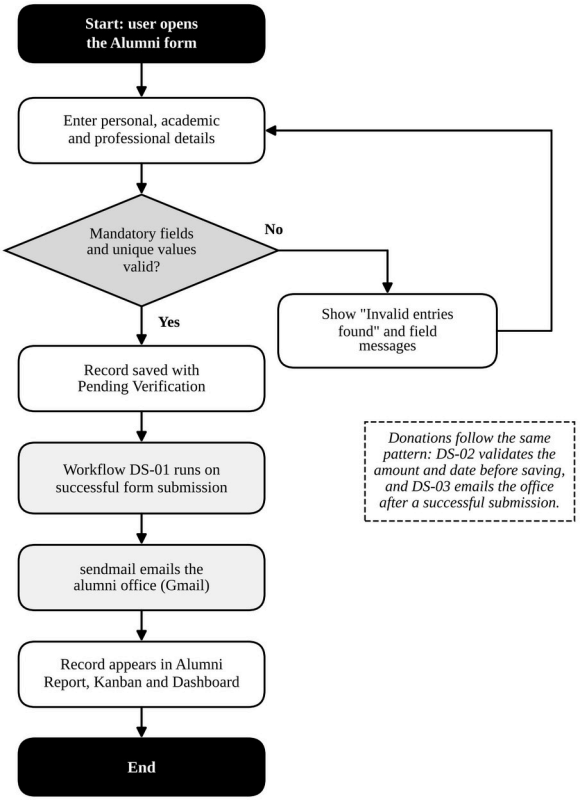


Figure 4.3: Alumni Registration and Notification Workflow

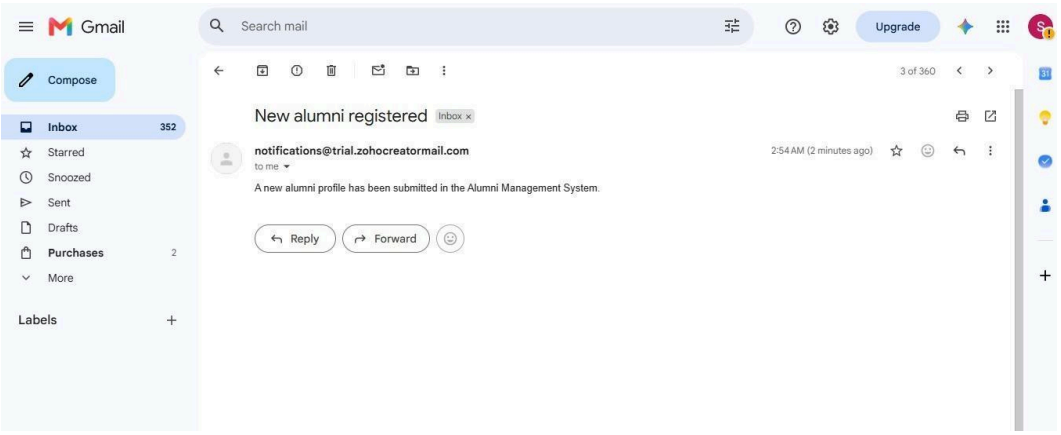
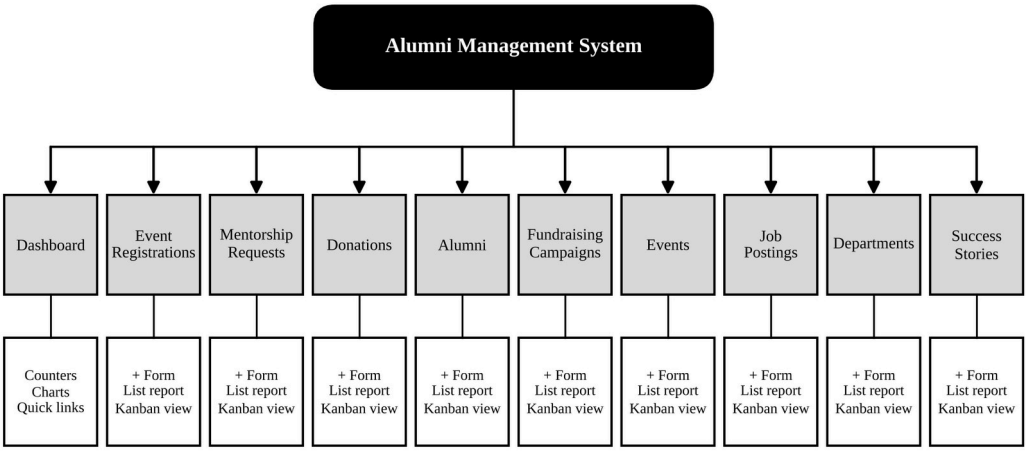


Figure 4.4: Alumni registration email notification received in the mailbox

4.5 User Interface and Navigation Design

The left navigation lists the Dashboard followed by one section per module, such as Event Registrations, Mentorship Requests, Donations, Alumni, Fundraising Campaigns, Events and Job Postings. Each section opens its form and reports. Quick links on the Dashboard open the most used forms.



Each module section opens its data-entry form, its list report (the Events report is a calendar) and a Kanban view grouped by a status or category field.

Figure 4.5: Application Navigation Structure

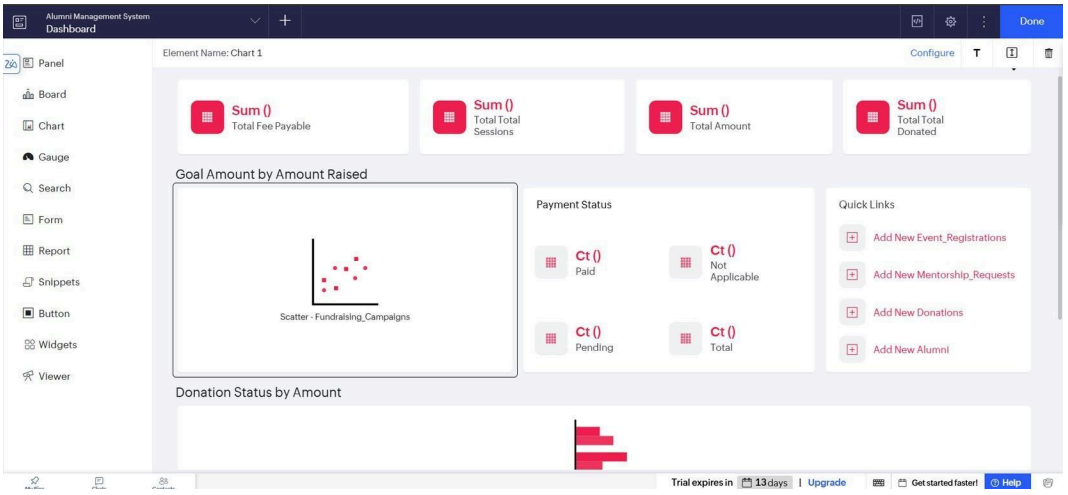


Figure 4.6: Dashboard page in the Zoho Creator page builder

4.6 Access Control Design

The intended permission matrix is shown below. A = Add, E = Edit, V = View, D = Delete, Own = own records only.

Table 4.11: Role–Permission Matrix

Component	Administrator	Coordinator	Finance Officer	Alumni	Student
Departments	A E V D	V	V	–	–
Alumni	A E V D	A E V	V	A, E Own	–
Events	A E V D	A E V D	V	V	V
Event Registrations	A E V D	A E V	V	A, V Own	–
Fundraising Campaigns	A E V D	V	A E V D	V	–
Donations	A E V D	V	A E V	A, V Own	–
Job Postings	A E V D	A E V	–	A, E Own	V
Mentorship Requests	A E V D	A E V	–	E Own	A, V Own
Success Stories	A E V D	A E V	V	A, V	V
Dashboard	V	V	V	–	–

CHAPTER 5
IMPLEMENTATION

This chapter reports the configuration visible in the application and its screenshots. Where the material does not demonstrate a feature, the report identifies that limit instead of assuming it was configured.

5.1 Application Setup

The application is named Alumni Management System and appears in the Applications workspace of the Zoho Creator account. The live navigation includes Dashboard, Event Registrations, Mentorship Requests, Donations, Alumni, Fundraising Campaigns, Events, Job Postings, Departments and Success Stories. Each section opens its form, its list or calendar report and a grouped (Kanban) view.

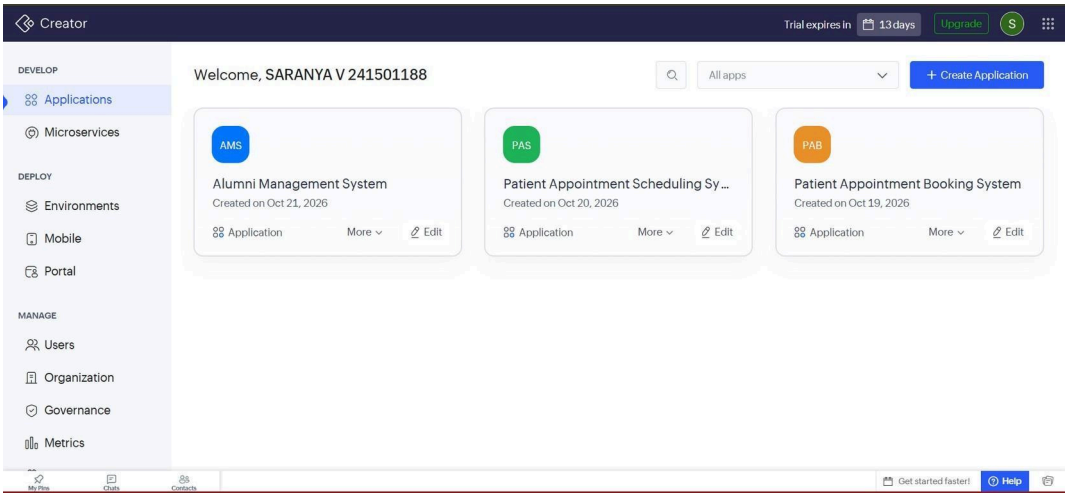


Figure 5.1: Alumni Management System in the Zoho Creator Applications workspace

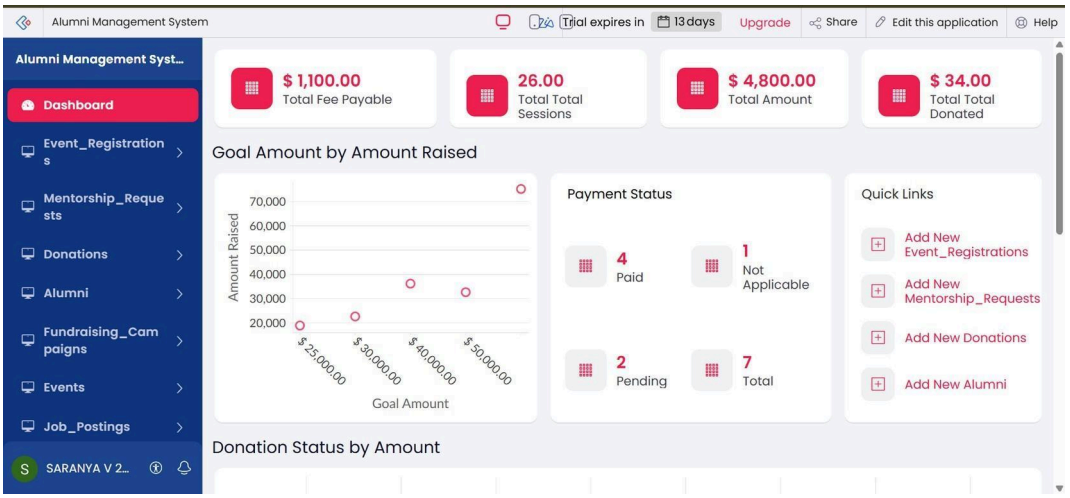


Figure 5.2: Live application home with the Dashboard and navigation menu

5.2 Forms and Data Capture

The Alumni form captures personal, academic, professional and engagement details including Full Name, Gender, Date of Birth, Email, Phone, Register Number, Department, Programme, Batch Start Year, Graduation Year and CGPA. The Phone field is encrypted and the Alumni ID field rejects duplicate values. Office-use fields such as Verification Status, Verified On, Profile Completeness, Active Mentees and Total Donated support the alumni office.

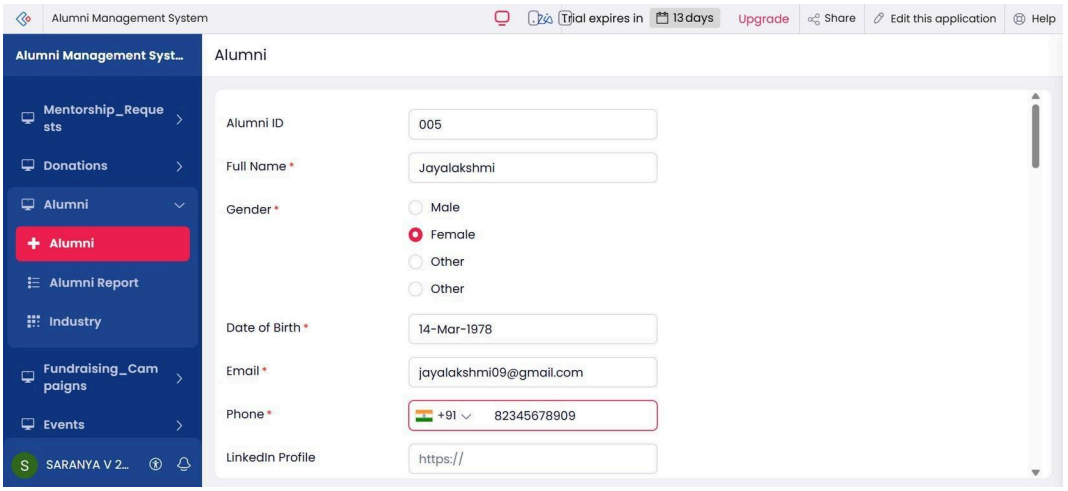


Figure 5.3: Alumni form in live mode – personal details

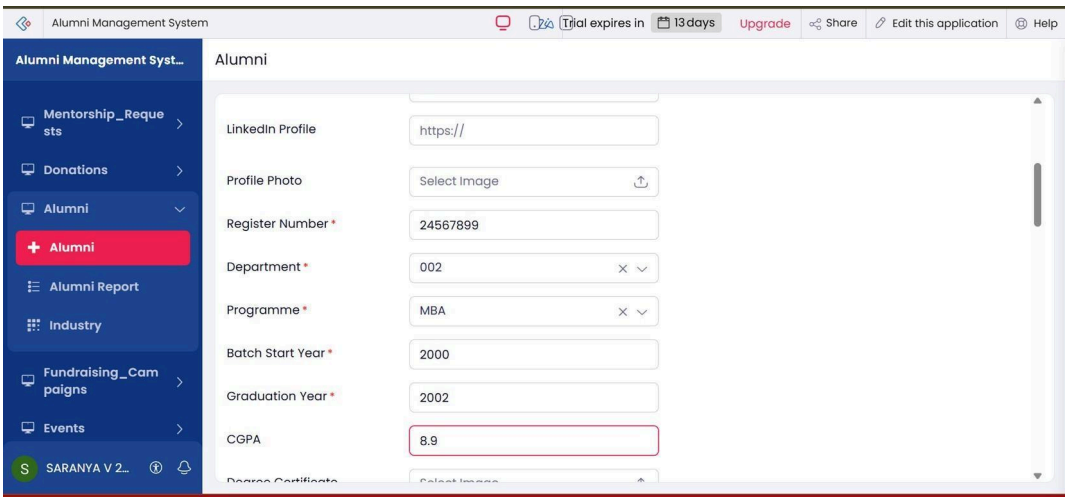


Figure 5.4: Alumni form – academic details with department lookup

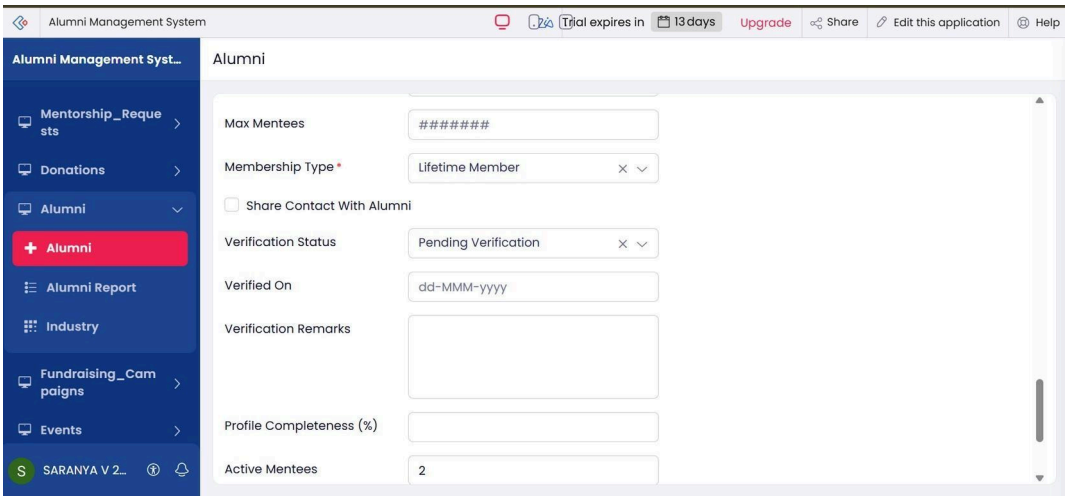


Figure 5.5: Alumni form – engagement and office-use fields

The Events form records the title, type, mode, dates, venue, capacity, registered count, seats available, registration deadline, fee and status of each event. The Donations form links a donation to an alumnus and a campaign and records the amount, date, payment mode, transaction reference, status, 80G receipt request, PAN number and anonymous-donation option.

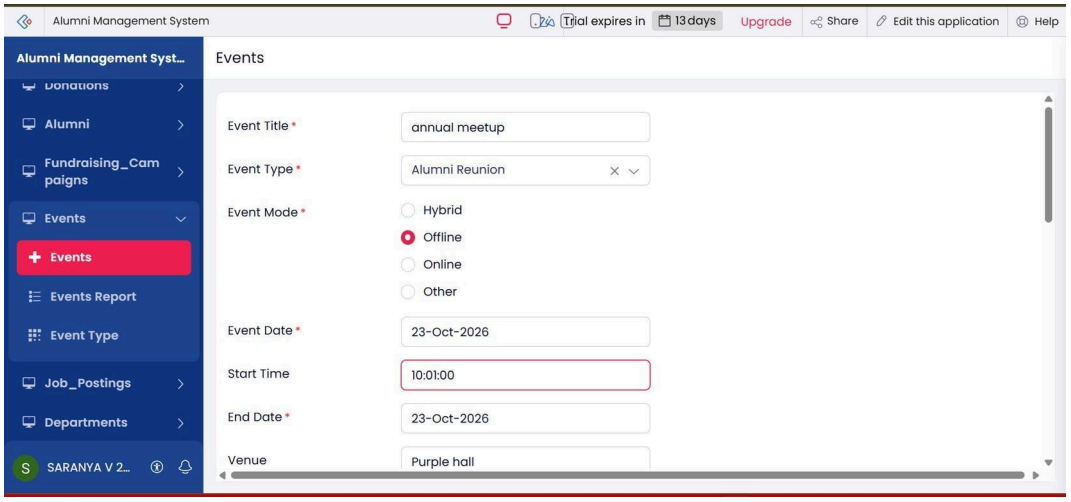


Figure 5.6: Events form with schedule, venue and type details

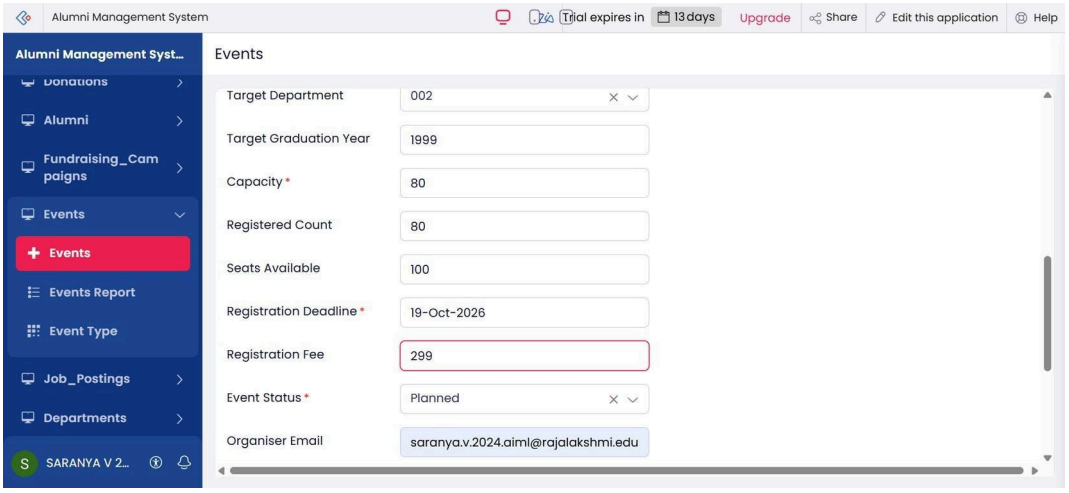


Figure 5.7: Events form – capacity, registration deadline, fee and status

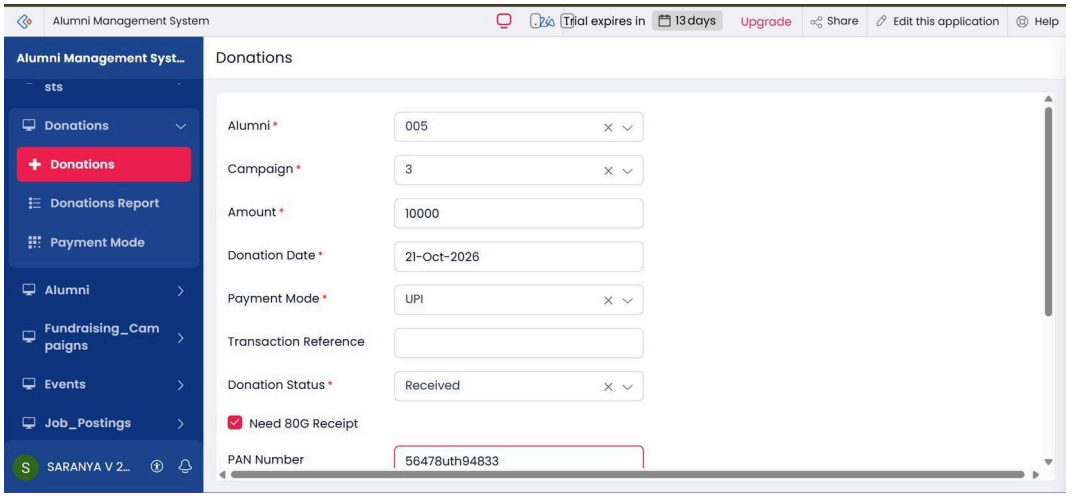


Figure 5.8: Donations form with alumni and campaign lookups

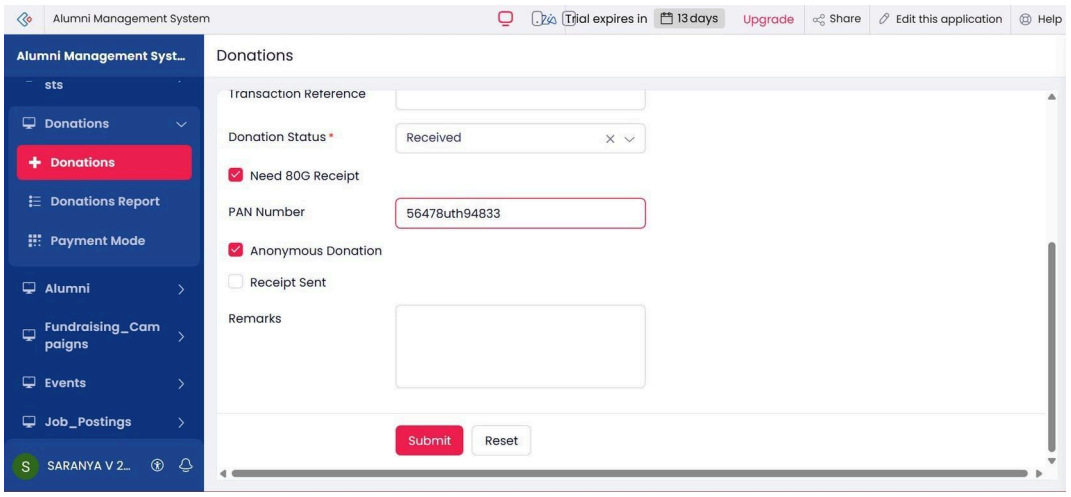


Figure 5.9: Donations form – status, 80G receipt and PAN fields

The Mentorship Requests form captures the student's details, the selected mentor, the mentorship area, the preferred mode and a message to the mentor. A Sessions subform

records each mentoring session, and the Total Sessions, Student Rating and Completion Notes fields close the request.

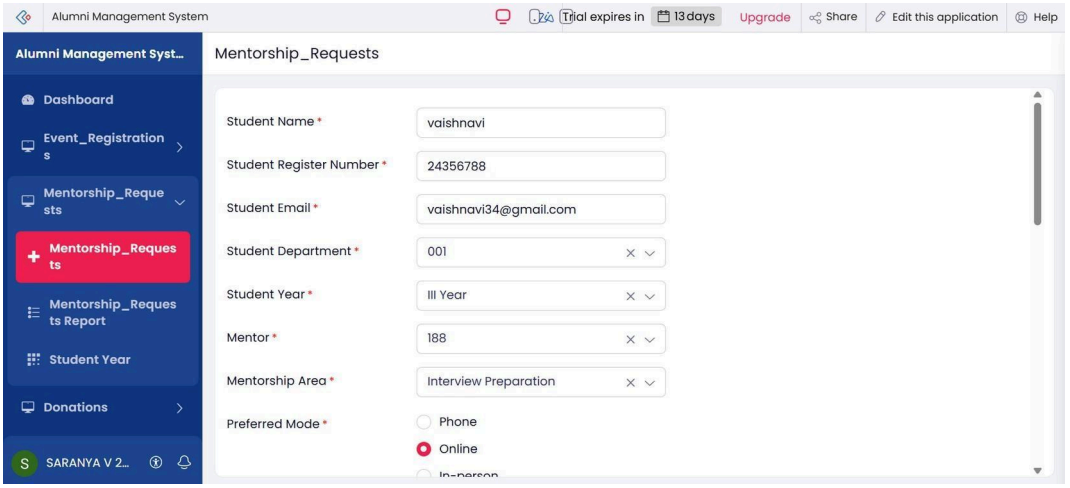


Figure 5.10: Mentorship Requests form with student and mentor details

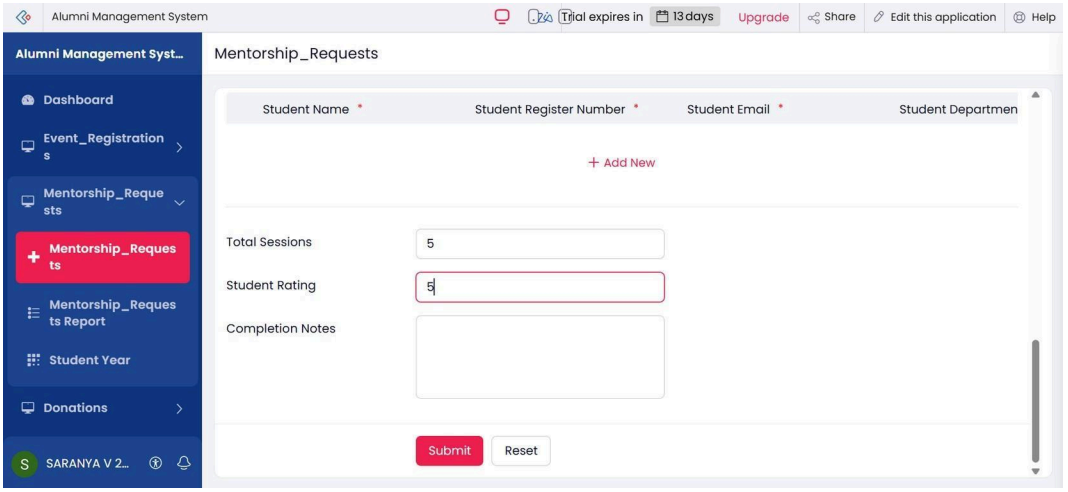


Figure 5.11: Mentorship Requests form with the Sessions subform

In the form builder, each field has a field name, a field link name used in Deluge, validation options such as mandatory and no-duplicate values, and data-privacy settings.

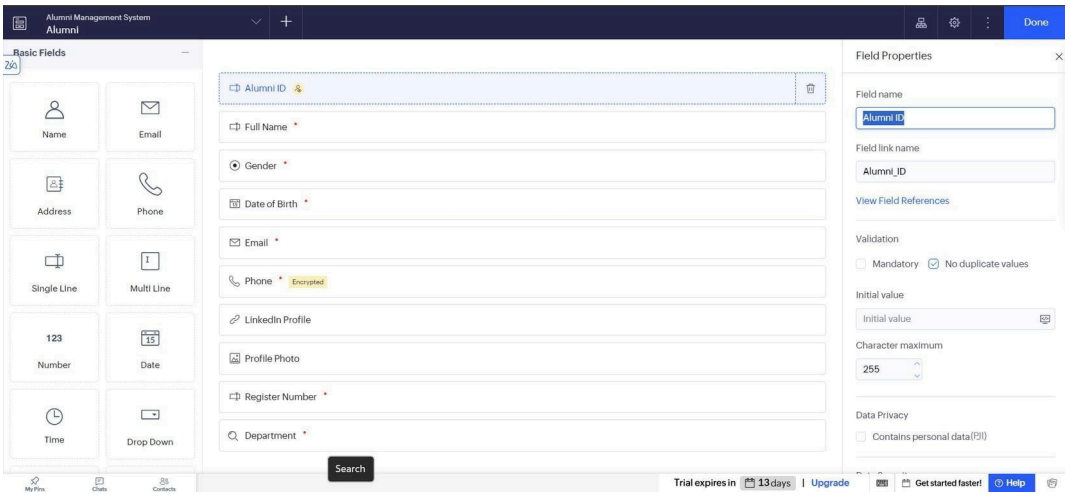


Figure 5.12: Alumni form in the builder with field properties

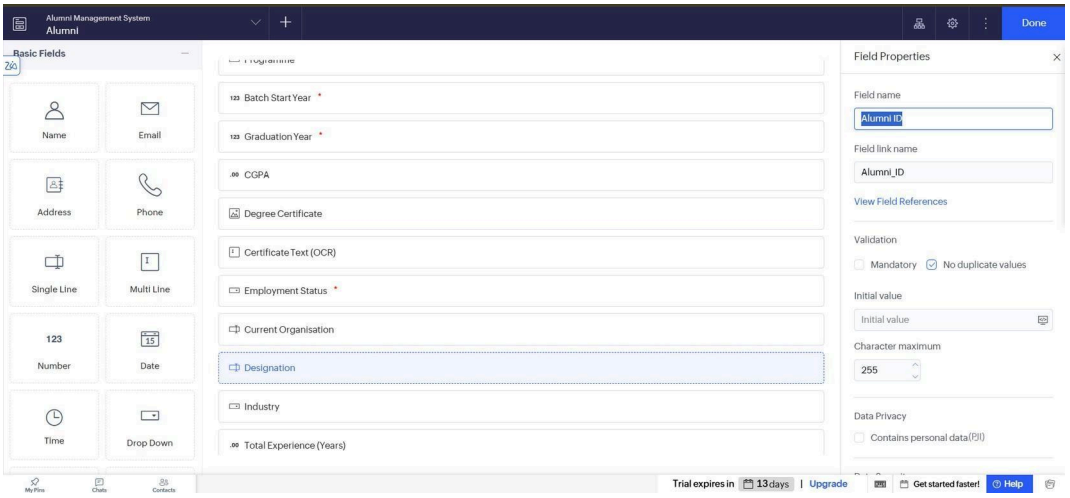


Figure 5.13: Alumni form builder – academic and professional fields

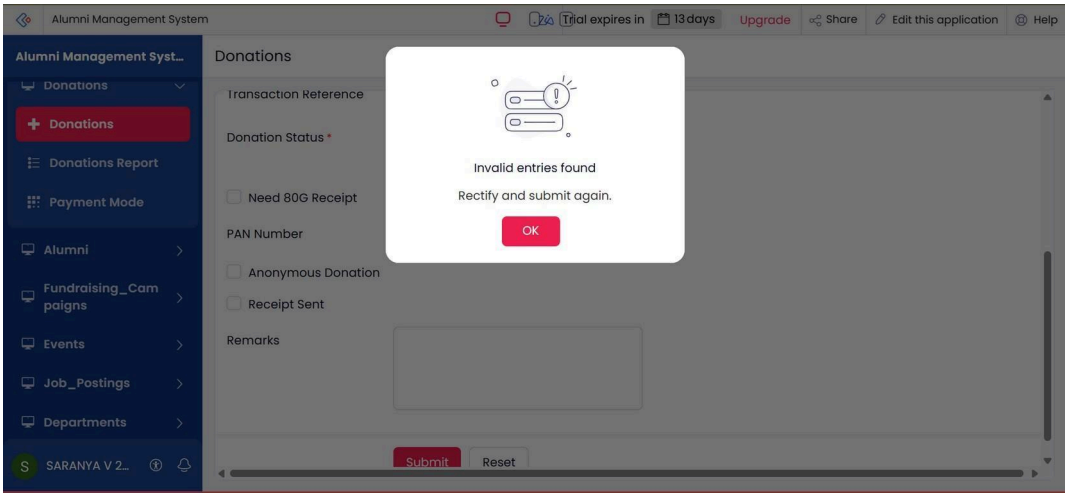


Figure 5.14: Donations form validation message on submission

5.3 Reports

Every form has a list report and a grouped Kanban view, and the Events report is displayed as a calendar. The reports show the sample records for alumni, events, donations, campaigns, job postings and success stories.

Table 5.1: Reports Implemented in the Application

Report Name	Type	Source Form	Grouping / Display	Purpose
Alumni Report	List	Alumni	All alumni records	Alumni directory
Industry	Kanban	Alumni	Grouped by industry	Alumni by sector
Events Report	Calendar	Events	Month, week and day views	Event schedule
Event Type	Kanban	Events	Grouped by event type	Event planning
Event_Registrations Report	List	Event_Registrations	Embedded in Dashboard	Registrations and feedback
Donations Report	List	Donations	All donations	Donation register
Payment Mode	Kanban	Donations	Grouped by payment mode	Payment tracking
Fundraising_Campaigns Report	List (grid)	Fundraising_Campaigns	Goal, raised, progress, donors	Campaign progress
Purpose	Kanban	Fundraising_Campaigns	Grouped by purpose	Campaign categories
Job_Postings Report	List	Job_Postings	Colour-coded job type	Career opportunities
Job Type	Kanban	Job_Postings	Grouped by job type	Job categories
Mentorship_Requests Report	List	Mentorship_Requests	All requests	Mentorship tracking
Student Year	Kanban	Mentorship_Requests	Grouped by student year	Mentorship demand
Success_Stories Report	List	Success_Stories	Colour-coded publish status	Story moderation
Category	Kanban	Success_Stories	Grouped by category	Story categories

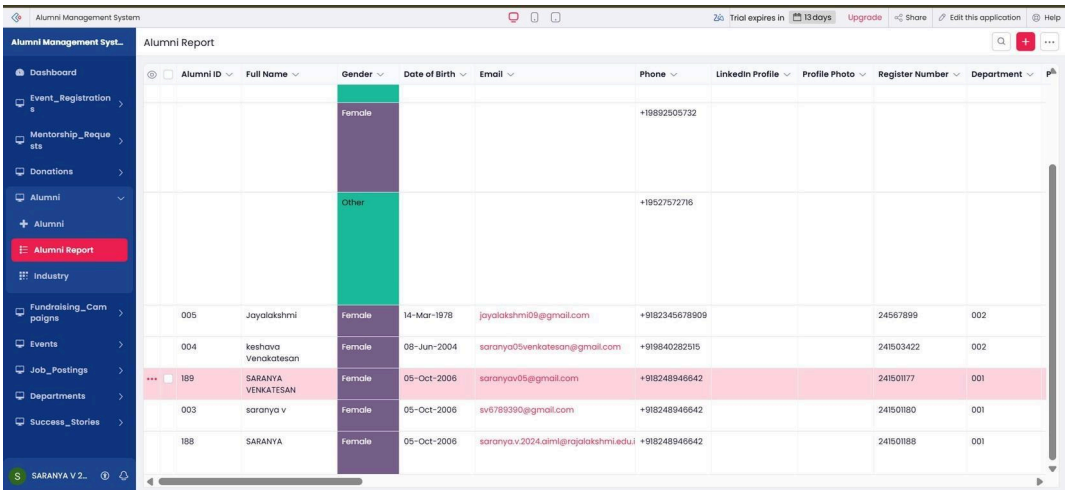


Figure 5.15: Alumni Report in live mode

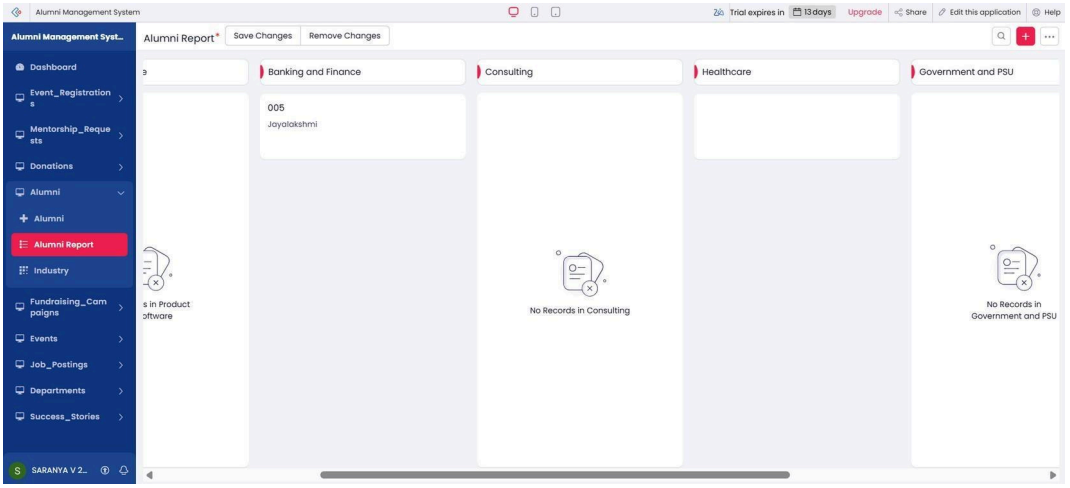


Figure 5.16: Alumni grouped by industry in a Kanban report

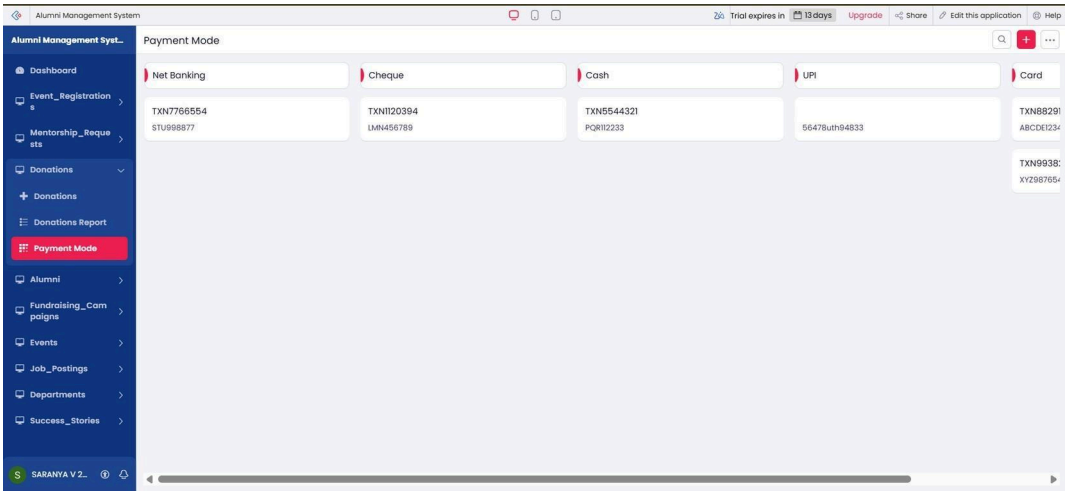


Figure 5.17: Donations grouped by payment mode in a Kanban report

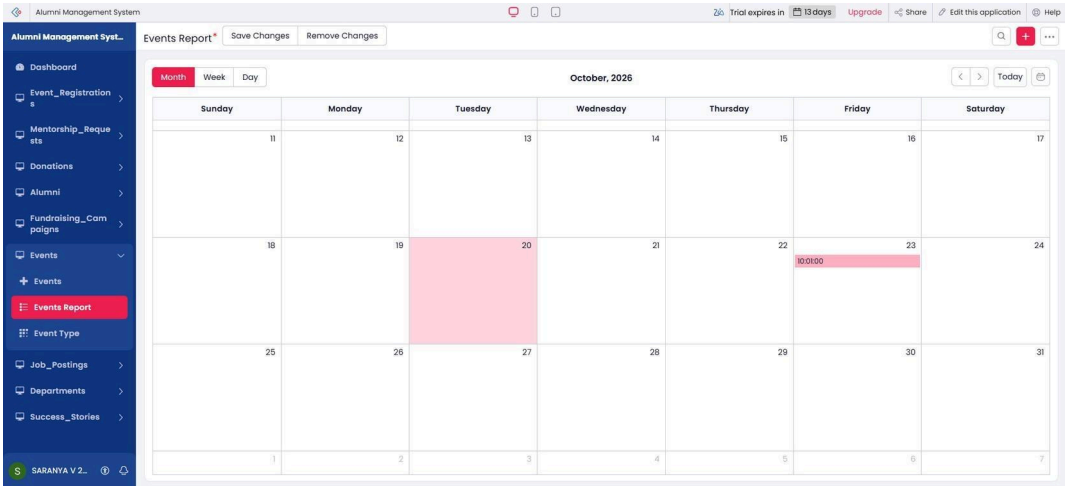


Figure 5.18: Events calendar report for October 2026

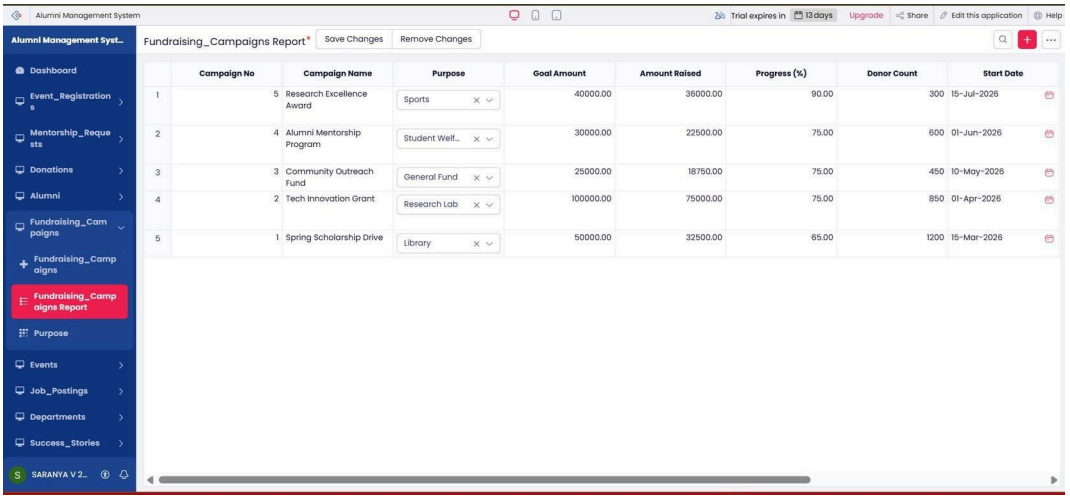


Figure 5.19: Fundraising Campaigns report with goal, amount raised and progress

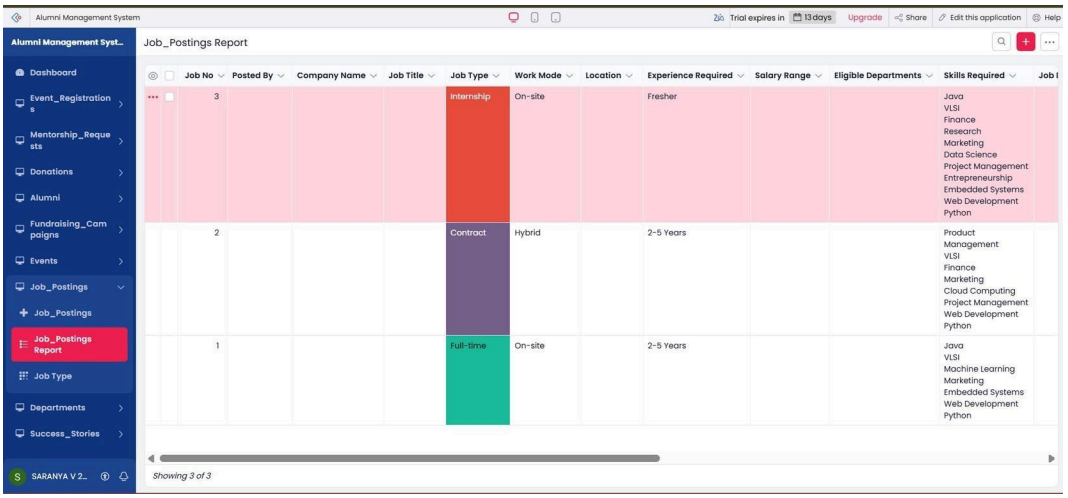


Figure 5.20: Job Postings report with job type and required skills

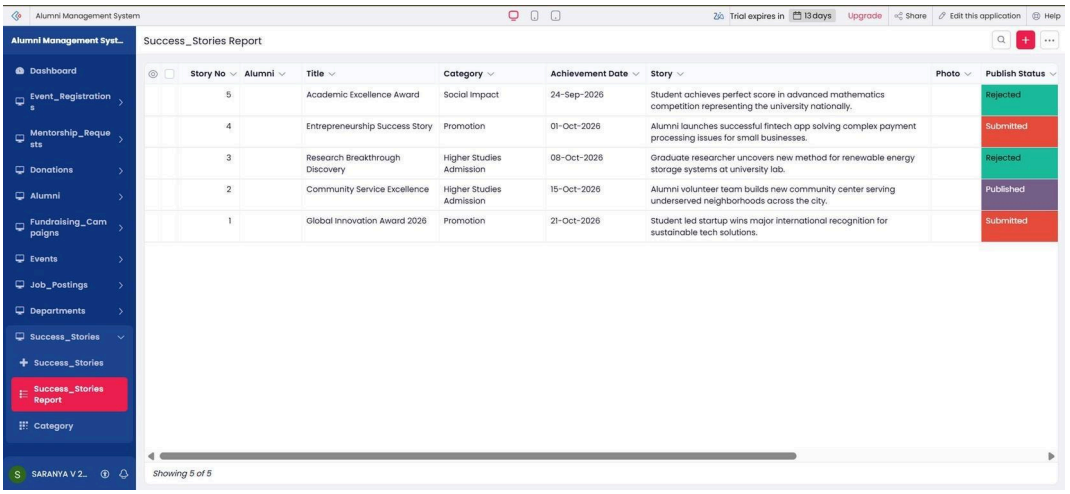


Figure 5.21: Success Stories report with publish status

5.4 Pages and Dashboards

The Dashboard page shows four summary counters (Total Fee Payable, Total Sessions, Total Amount and Total Donated), a chart of campaign goal amount against amount raised, payment-status counters (Paid, Pending, Not Applicable and Total), quick links to add event registrations, mentorship requests, donations and alumni, a Donation Status by Amount chart and an embedded Event Registrations report.

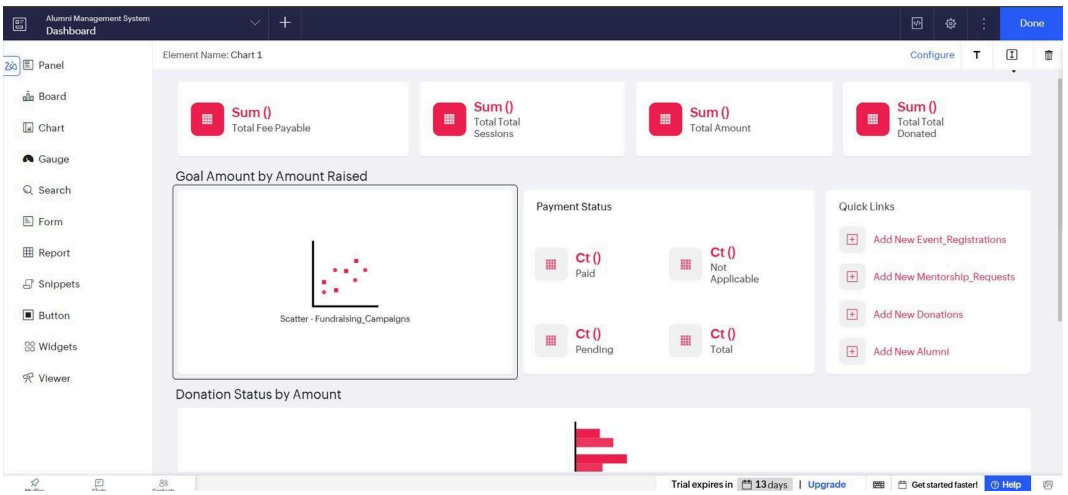


Figure 5.22: Dashboard page in the Zoho Creator page builder

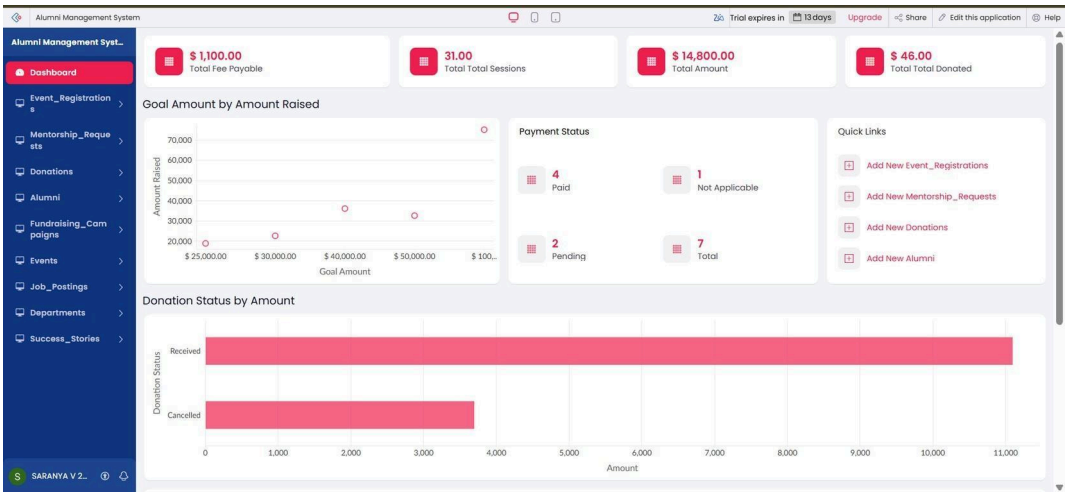


Figure 5.23: Live Dashboard with counters, charts and quick links

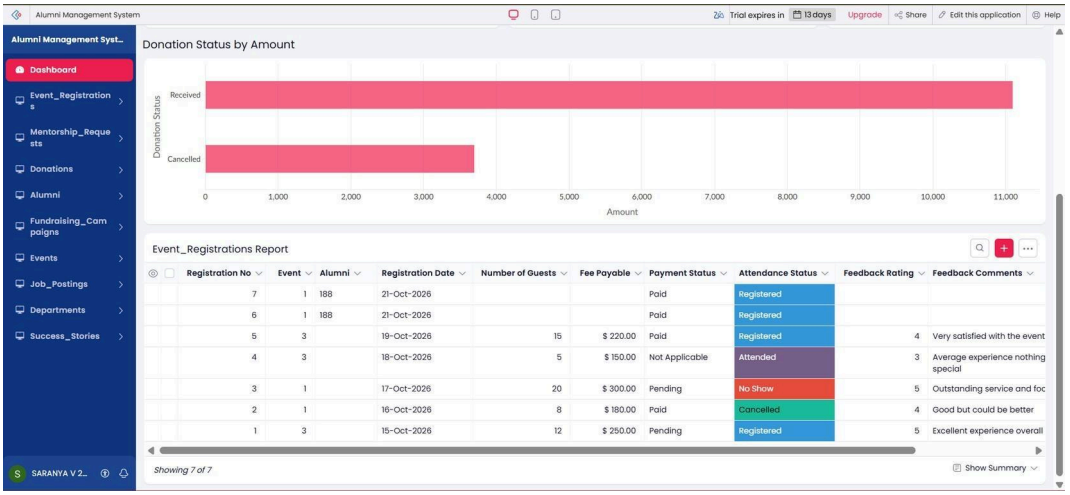


Figure 5.24: Dashboard with the Donation Status chart and embedded Event Registrations report

5.5 Deluge Scripting

Deluge scripts implement the notification and validation rules. The registration email and the donation notification are on-success scripts; the donation rules run on validation and cancel the submission when a value is invalid. The complete scripts appear in Appendix C.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Alumni → Created → Successful form submission	Email the alumni office when a new alumnus registers.	sendmail task with from, to, subject and message
DS-02	Donations → Created or Edited → Validations on form submission	Reject donations below Rs. 100 and donation dates in the future.	Conditions, alert, cancel submit, zoho.currentdate
DS-03	Donations → Created → Successful form submission	Email the alumni office when a donation is recorded.	Input fields, sendmail

Code 5.1: Alumni registration email notification (DS-01)

```
// DS-01 | Alumni | Created | Successful form submission
sendmail
[
    from :zoho.adminuserid
    to :"saranyav05@gmail.com"
    subject :"New alumni registered"
    message :"A new alumni profile has been submitted in the Alumni
Management System."
]
```

5.6 Workflows and Automation

Form workflows run the Deluge scripts when records are submitted. The Alumni_Registered_Email workflow was tested by submitting a new Alumni record; the notification was delivered to the alumni office mailbox from the Zoho Creator notification address within a minute. A blueprint named mentorship_lifecycle is enabled on the Alumni form.

Table 5.3: Workflows and Blueprint Processes

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Alumni_Registered_Email	Form workflow	Alumni created; successful form submission	Every new record	Email alumni office	DS-01
Donation_Validation	Form workflow	Donations created or edited; validations on submission	Amount < 100 or future date	Alert and cancel submit	DS-02
Donation_Notification	Form workflow	Donations created; successful form submission	Every new record	Email alumni office	DS-03
mentorship_lifecycle	Blueprint	Alumni record lifecycle	Starts at Stage1	user flow (Stage1 → Stage2), admin flow (Stage3 → Stage1)	Blueprint stages

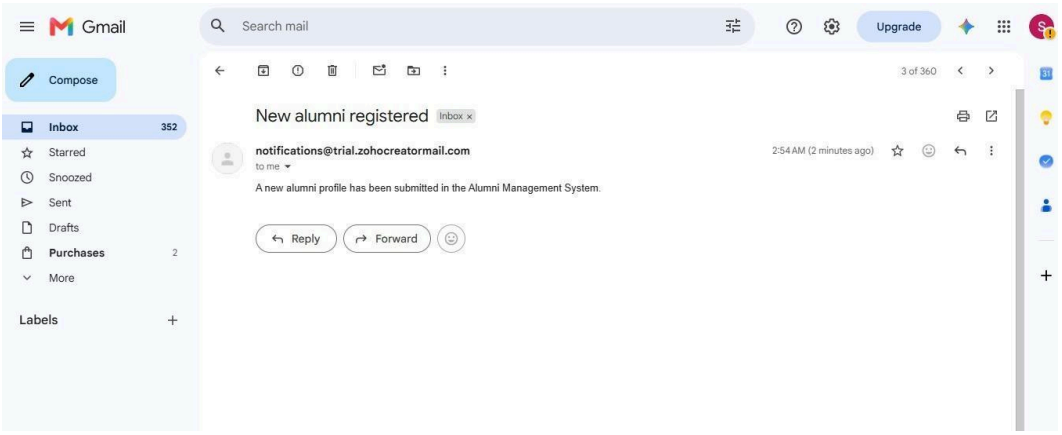


Figure 5.25: Alumni registration email notification received in Gmail

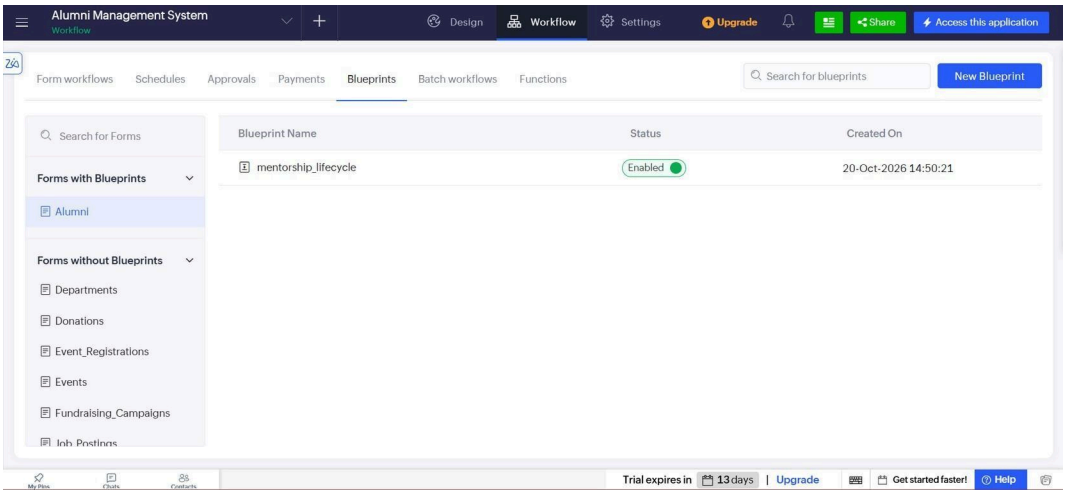


Figure 5.26: Blueprints page showing the enabled mentorship_lifecycle blueprint

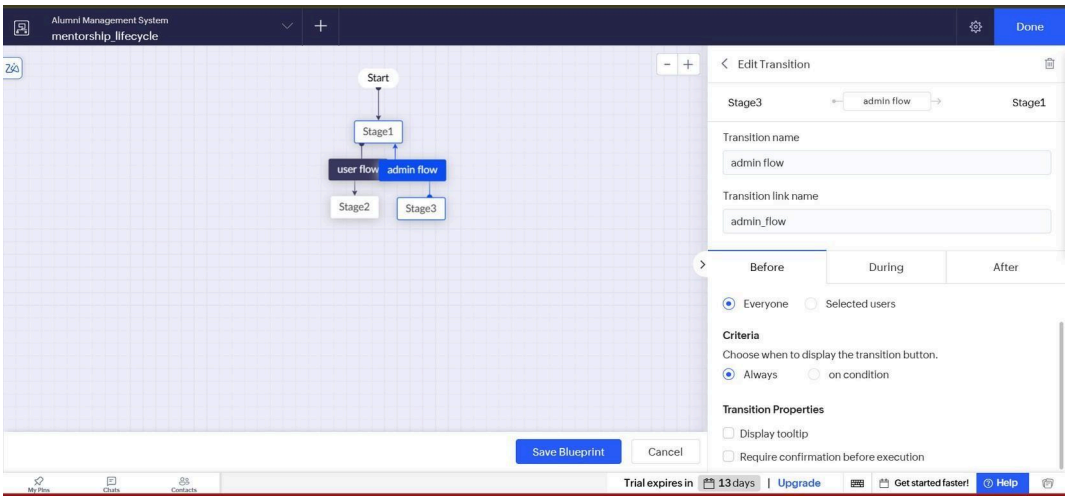


Figure 5.27: mentorship_lifecycle blueprint with stages and transitions

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service	Method	Purpose and Evidence
Email notification	Zoho Creator mail	Deluge sendmail	Registration email received in Gmail
Team messaging	Zoho Cliq	Creator connection	Configured for announcements; delivery not verified
Calendar	Zoho Calendar	Creator connection	Configured for event dates; sync not verified

5.8 User Management and Role-Based Access

The User Permissions page lists the Alumni Coordinator and Finance Officer permission profiles with descriptions of their access, together with the default Read and Write profiles. Portal User Permissions are available for alumni and student portal users. Separate sign-in tests for each profile remain to be demonstrated.

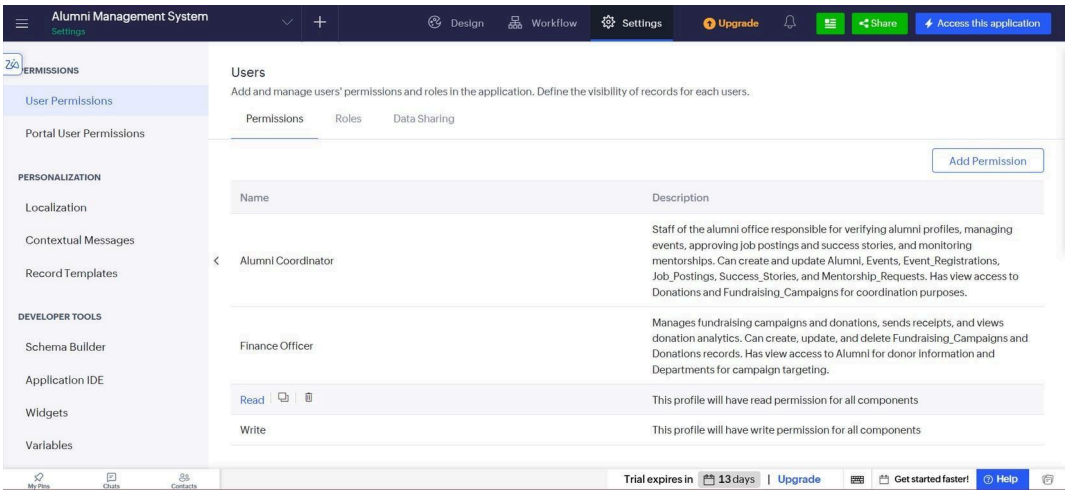


Figure 5.28: User Permissions page with the Alumni Coordinator and Finance Officer profiles

5.9 AI and Analytics Features

The Dashboard provides native analytics through counters and charts of donations, payment status and campaign progress. Zoho Creator's AI-assisted builder was used during development to suggest and generate application components, which were then reviewed and corrected. No live Zia prediction, OCR or sentiment field is claimed in this report.

5.10 Pro-Code Extension

No Catalyst service, widget or external pro-code component is deployed. Deluge scripts form the scripting layer of the application.

5.11 Mobile and Portal Access

Browser access was verified. The Zoho Creator native mobile app and a customer portal for alumni and students were not exercised, and no portal configuration is claimed.

CHAPTER 6
TESTING AND VALIDATION

6.1 Testing Strategy

Verification combined inspection of the forms, reports and workflows in the builder, submissions in the live application and confirmation of email delivery. Tests that were not executed are marked as such rather than claimed.

6.2 Test Cases

Table 6.1: Functional and Workflow Test Cases

TC ID	FR	Test	Input / Method	Expected	Status
TC-01	FR-01	Submit Alumni form with valid data	Live form	Record saved	Pass
TC-02	FR-01	Submit Alumni form with empty mandatory fields	Live form	Mandatory-field messages shown	Pass
TC-03	FR-13	Registration email after Alumni submission	DS-01	Email received	Pass
TC-04	FR-07	Donation amount below 100	DS-02	Submission blocked	Pass
TC-05	FR-07	Donation date in the future	DS-02	Submission blocked	Pass
TC-06	FR-13	Donation notification email	DS-03	Email received	Pass
TC-07	FR-11	Reports show sample records	Live reports	Records listed	Pass
TC-08	FR-12	Dashboard counters and charts	Live Dashboard	Values displayed	Pass
TC-11	FR-11	Events calendar report	Live report	Event shown on its date	Pass
TC-09	FR-09	Blueprint enabled with stages	Blueprints page	Blueprint enabled	Pass
TC-10	FR-14	Separate-role access	Role sign-in	Only permitted modules	Not run

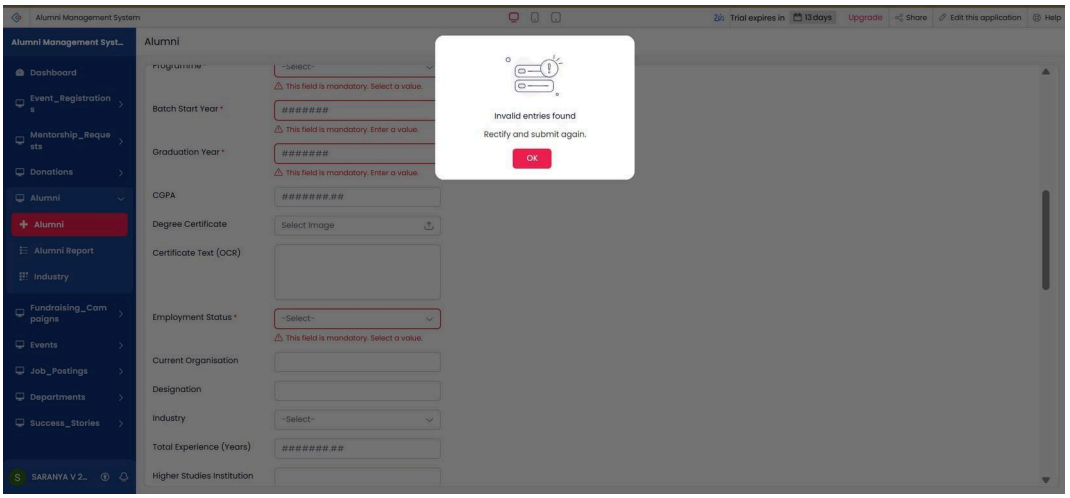


Figure 6.1: Empty-submit validation errors in the Alumni form

6.3 Debugging Deluge Scripts

Scripts were saved in the Deluge editor, which reports unknown field names and syntax errors before saving. Field link names were checked against the form builder so that input.Full_Name, input.Email, input.Amount and input.Donation_Date match the forms. Each workflow was then tested by submitting a record in the live application and checking the mailbox or the validation alert.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Cause	Correction	Status
BUG-01	Amount fields displayed in USD	Default currency setting	Set currency fields to INR	Open
BUG-02	Dashboard shows generic counters	Default dashboard layout	Revise counters to alumni KPIs	Open
BUG-03	Script error on unknown field name	Link name differed from label	Used the field link name	Closed
BUG-04	Email not visible in inbox	Delivered to spam folder	Checked spam folder	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role	Feedback	Action
Not collected	Alumni Coordinator	No genuine UAT response available	Schedule a guided demonstration
Not collected	Alumni / Student	No genuine UAT response available	Record feedback after a role demo

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application uses the Zoho Creator multi-stage deployment model of Development, Stage and Production. It was added to Environments on 21 October 2026, and version
1.0 was published to Stage for testing and then to Production, where end users access it through the authenticated live URL listed in the Project at a Glance.

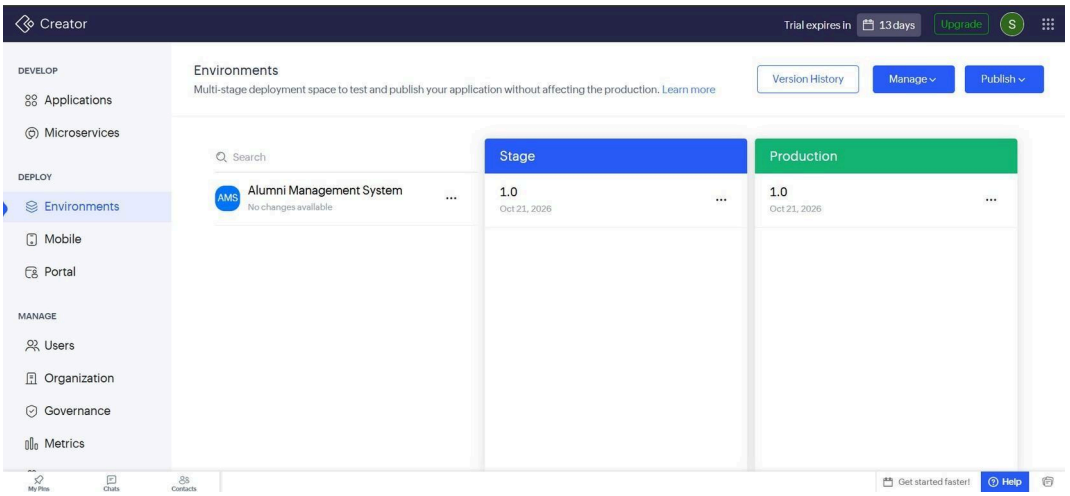


Figure 7.1: Environments page with version 1.0 published to Stage and Production

Further changes are made only in the Development environment and released through the Publish menu, first to Stage and then to Production. The menu also shows the number of Production publishes used in the account.

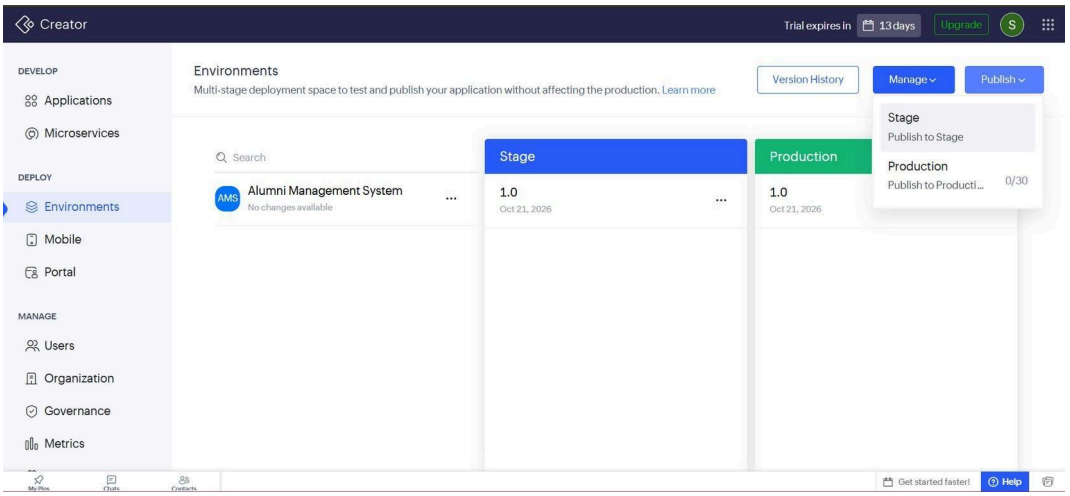


Figure 7.2: Publish menu with the Publish to Stage and Publish to Production options

Records do not move between environments during publishing. The Development environment therefore opens with empty counters and charts, while the records entered before deployment remain with the live application. This separation lets changes be tested without affecting live data.

No.	Deployment Item	Status
1	All tests executed in Development or Stage	Functional tests executed; Stage version 1.0 reviewed
2	Sample data removed or isolated	Sample data retained for demonstration
3	Roles and users tested in live app	Permission profiles defined; role tests pending
4	Business-service connections authorised	Zoho Cliq and Calendar connections configured
5	Workflows executed	Registration and donation workflows tested
6	Authenticated live URL opened	Verified 21 October 2026
7	Application published to Stage and Production	Version 1.0 published on 21 October 2026

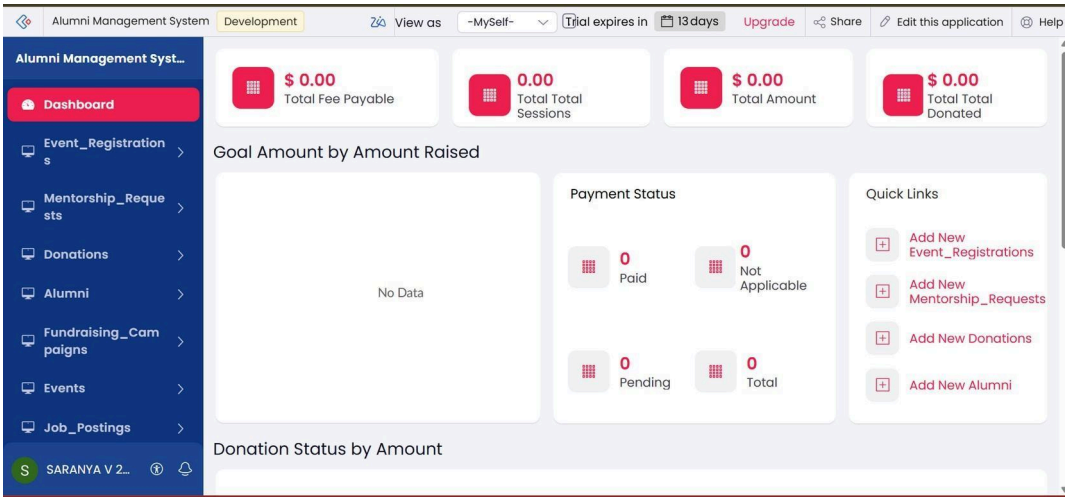


Figure 7.3: Development environment of the application with no records

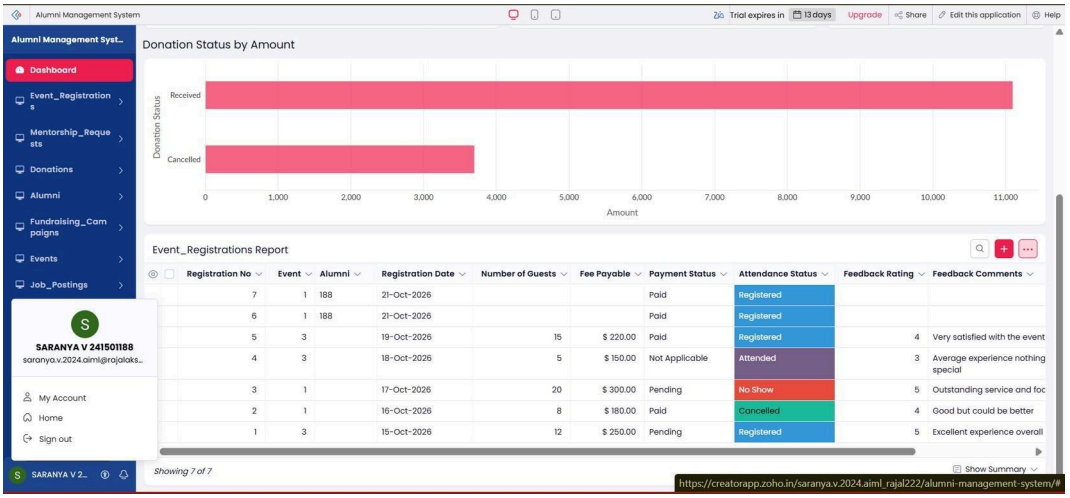


Figure 7.4: Authenticated live application with the signed-in user and application

URL Table 7.1: Deployment Checklist

7.2 Performance Optimisation

Lookup fields avoid re-entering alumni and department details, and filtered and grouped reports reduce the records shown for daily tasks. These are implementation choices, not measured performance improvements.

Table 7.2: Performance Optimisation Results

Technique	Component	Before	After	Measured Result
Shared lookup values	Alumni and Department lookups	Manual re-entry	Related records	Consistency observed; effort not measured
Grouped reports	Kanban boards	All records view	Grouped by status	Grouping verified; latency not measured
Validation before save	Donations (DS-02)	Invalid values saved	Submission blocked	Verified by test

7.3 Security Features

Table 7.3: Security Controls Implemented

Control	Applied Location	Evidence and Limit
Authenticated access	Creator application URL	Sign-in required; verified
Role-based permissions	Five roles	Defined; separate sign-in tests pending
Input validation	Mandatory fields and DS-02	Verified by submission tests
Sensitive fields	Phone, PAN and email fields	Field privacy settings to review

7.4 Governance and Administration

The application owner manages users, roles and settings. Changes to workflows are recorded in this report's defect log, and the Deluge source is retained in Appendix C. A native audit trail configuration was not verified.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application is live with nine linked forms and sample records. Submitting the Alumni form saves the record with Pending Verification status and sends an email to the alumni office. The Dashboard shows a total amount of 14,800, 31 mentorship sessions, registration payment status (Paid 4, Pending 2, Not Applicable 1, Total 7), a campaign goal-versus-raised chart, a donation-status chart and the embedded registrations report. Five fundraising campaigns, seven event registrations, three job postings and five success stories are recorded.

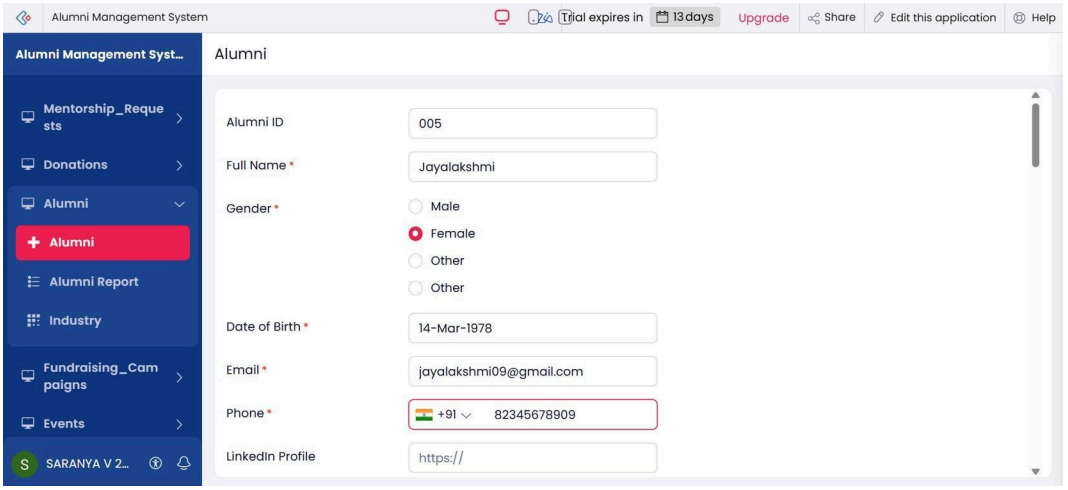


Figure 8.1: Alumni form in live mode

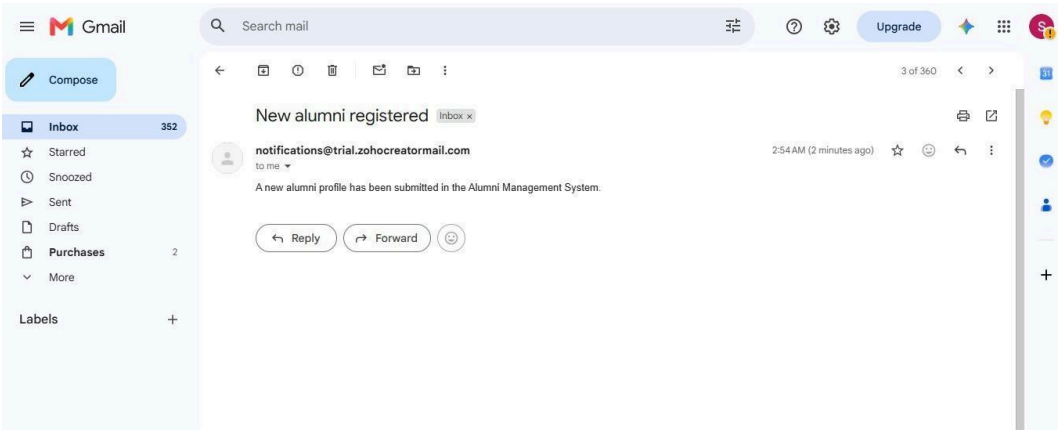


Figure 8.2: Alumni registration email notification

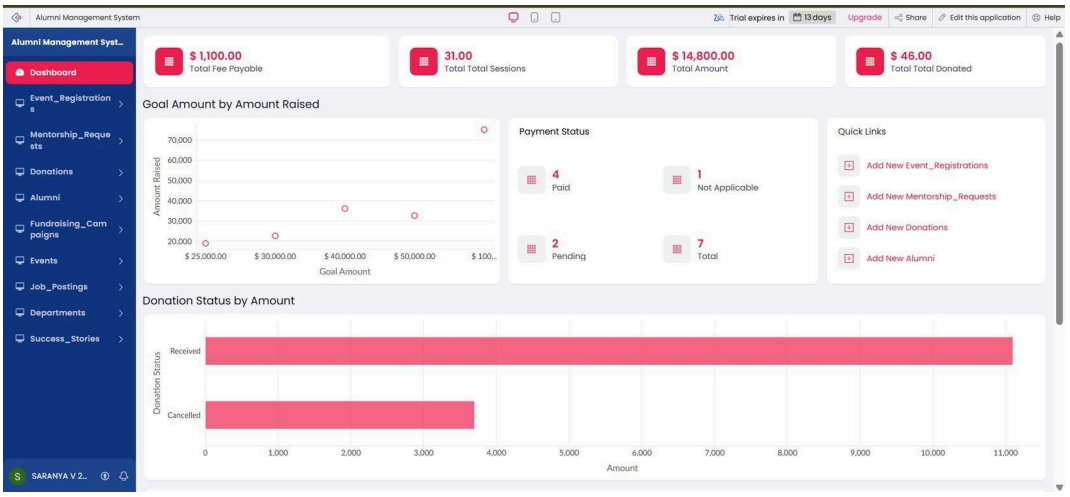


Figure 8.3: Live Dashboard with donation and campaign charts

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Implementation	Outcome / Limit
Structured forms	Nine linked forms	Forms and lookups available with sample data
Status visibility	Status fields, Kanban reports, calendar, blueprint	Records visible by status and date
Reports and Dashboard	List, calendar and Kanban reports; Dashboard	Counters, charts and embedded report displayed
Automation	DS-01 to DS-03 workflows	Email and validation verified
Access control	Permission profiles	Defined; role tests pending
Deployment	Development, Stage and Production environments	Version 1.0 published to Stage and Production

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Manual Process	Proposed Application
Alumni data	Spreadsheets, unverified, duplicated	One Alumni form with verification status
Notifications	Manual messages	Automatic email on registration and donation
Events and donations	Separate forms and ledgers	Linked records, reports and dashboard counters
Jobs and mentorship	Informal messages	Status boards and mentorship blueprint
Time and cost	No baseline measured	No percentage saving claimed

8.4 Limitations of the Low-Code Approach

The trial account limits availability. Some configuration, such as the currency of amount fields and the dashboard counters, needed review after setup. Builder capabilities constrain layout choices, and platform-specific definitions create a migration cost. No production-scale performance claim is made.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Field definitions, lookups, report layouts and the Dashboard lie toward the no-code end. Deluge validation and notification scripts form the low-code portion. No pro-code component was required; a move to custom services would be justified only by measured integration or processing needs.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Alumni Management System is deployed in Zoho Creator with nine related forms, list, calendar and Kanban reports, a Dashboard page, a blueprint, Deluge workflows and permission profiles. Alumni registration triggers an email to the alumni office, donation rules prevent invalid entries, and version 1.0 has been published through Stage to Production. Role-isolation tests and user feedback remain to be documented.

9.2 Future Enhancements

- Set all amount fields to INR and extend the Dashboard with alumni KPIs such as verified alumni, upcoming events and active mentorships.
- Add a customer portal for alumni and students and test each role with separate users.
- Add scheduled reminders for events and job expiry, and evaluate payment-gateway collection for donations.

9.3 Learning Outcomes and Career Relevance

The project developed skills in data modelling, form and report design, Deluge scripting, workflow and blueprint configuration and evidence-based testing. These apply to low-code development, automation engineering and solutions consulting.

REFERENCES

References are listed in IEEE style and cited in the report.

[1] D. Gao, F. Fagerholm, and V. Toivanen, “What does current research say about the viability of low-code development? A systematic literature review,” Journal of Systems and Software, vol. 239, art. 112893, 2026, doi: 10.1016/j.jss.2026.112893.

[2] M. O. Ajimati, N. Carroll, and M. Maher, “Adoption of low-code and no-code development: A systematic literature review and future research agenda,” Journal of Systems and Software, vol. 222, art. 112300, 2025, doi: 10.1016/j.jss.2024.112300.

[3] Zoho Corporation, “A Quickstart Guide to Zoho Creator,” Zoho Creator Help. [Online]. Available: https://www.zoho.com/creator/help/new-quickstart-guide.html. Accessed: 21 Oct. 2026.

[4] Zoho Corporation, “Understanding workflows,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/understand-workflows/articles/understand-workflows. Accessed: 21 Oct. 2026.

[5] Zoho Corporation, “Creating blueprints,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/blueprints/articles/create-a-blueprint. Accessed: 21 Oct. 2026.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

This checklist distinguishes features evidenced in the application and screenshots from features not verified.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Yes	Dashboard and module sections	5.1	Fig. 5.2
II	Forms with ≥ 8 distinct field types	Yes	Nine forms; multiple field types	4.3, 5.2	Tables 4.2–4.10
II	Lookup field (relationship between forms)	Yes	Department, Alumni, Event, Campaign lookups	4.2	Table 4.1
II	Subform (parent–child data)	Yes	Employment History and Sessions subforms	4.3, 5.2	Tables 4.3, 4.9
II	Field validations / layout rules	Yes	Mandatory fields; DS-02	5.5, 6.2	Fig. 5.14
II	List report with filter, grouping, custom action	Partial	List reports with grouping and colour coding	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	Yes	Kanban views and Events calendar	5.3	Figs. 5.16, 5.18
II	Pivot table / chart report	Partial	Dashboard charts	5.4	Fig. 5.23
II	Page / dashboard built with UI Builder	Yes	Dashboard page	5.4	Fig. 5.23
II	Deluge – form events	Yes	DS-01 to DS-03	5.5	App. C
II	Email / in-app notifications	Yes	Registration and donation emails	5.6	Fig. 4.4
II	Form workflow	Yes	Three form workflows	5.6	Table 5.3
II	Scheduled workflow	No	Not implemented	–	–
II	Approval process	Partial	Status-based review	5.6	–
II	Blueprint	Yes	mentorship_lifecycle (enabled)	5.6	Figs. 5.26, 5.27
II	Batch / report workflow	No	Not implemented	–	–
II	Testing & debugging evidence	Yes	Test cases and defect log	6	Table 6.1
III	Integration with a Zoho service	Partial	Zoho Cliq and Calendar connections	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	No	Not implemented	–	–

III	Users, roles and permission profiles	Partial	Coordinator and Finance Officer profiles	5.8	Fig. 5.28

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Zia / AI feature	Partial	AI-assisted builder used in development	5.9	–
III	Analytics (charts)	Yes	Dashboard charts	5.4	Fig. 5.23
IV	Performance optimisation applied & measured	Partial	Lookups and grouped reports	7.2	Table 7.2
IV	Deployment across environments	Yes	Version 1.0 in Stage and Production	7.1	Figs. 7.1, 7.2
IV	Security controls	Partial	Authentication, validation, roles	7.3	Table 7.3
IV	Governance (audit trail, administration)	Partial	Owner administration	7.4	–
V	Pro-code extension	No	Not required	5.10	–
V	Low-code limitations & spectrum analysis	Yes	Discussion	8.4, 8.5	–
–	Mobile app / portal access	No	Browser only	5.11	–

Coverage summary: 15 evidenced items, 9 partial items and 5 absent or unverified items out of 29.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture and business value of low-code application development using Zoho Creator.	Ch. 1, 2	Review by student	
CO2	Develop business applications using Zoho Creator forms, reports, workflows and Deluge scripting.	Ch. 4, 5, 6	Review by student	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections and role-based security.	Ch. 4.6, 5.7–5.8	Review by student	
CO4	Deploy, secure and manage Zoho Creator applications following governance best practices.	Ch. 7	Review by student	
CO5	Evaluate low-code development strategies, AI-assisted application development and the transition to pro-code.	Ch. 5.9–5.10, 8.4–8.5	Review by student	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

The Deluge source below implements the workflow logic described in Chapter 5. Each script is reproduced as configured in the application.

C.1 Alumni Workflow – Registration Email Notification

```
// DS-01 | Alumni | Created | Successful form submission
sendmail
[
    from :zoho.adminuserid
    to :"saranyav05@gmail.com"
    subject :"New alumni registered"
    message :"A new alumni profile has been submitted in the Alumni
Management System."
]
```

C.2 Donations Workflow – Amount and Date Validation

```
// DS-02 | Donations | Created or Edited | Validations on
form submission
if(input.Amount < 100)
{
    alert "Minimum donation amount is Rs. 100.";
    cancel submit;
}
if(input.Donation_Date > zoho.currentdate)
{
    alert "Donation date cannot be in the future.";
    cancel submit;
}
```

C.3 Donations Workflow – Donation Notification Email

```
// DS-03 | Donations | Created | Successful form
submission sendmail
[
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"Donation received - Rs. " + input.Amount
    message :"A donation of Rs. " + input.Amount + " dated " +
input.Donation_Date + " has been recorded in the Alumni Management
System.<br><br>Regards,<br>Alumni Management System"
]
```

APPENDIX D – USER MANUAL

This user guide describes the main web tasks in the application. Access depends on a Creator user account and permissions assigned by the application owner.

D.1 Accessing the Application

Open the Alumni Management System in Zoho Creator using the URL provided by the application owner and sign in with your assigned account. Use the left navigation to open the Dashboard, Alumni, Events, Event Registrations, Fundraising Campaigns, Donations, Job Postings, Mentorship Requests or Success Stories. The screenshots document web access; mobile and portal steps are not verified.

D.2 Instructions for Administrator and Alumni Coordinator

- 1. Open Alumni and choose Add to register an alumnus. Enter personal, academic and professional details, select the department and submit. The alumni office receives an email notification.
- 2. Use the Kanban boards to review events, job postings and mentorship requests by status, and update the status of each record as it progresses.

D.3 Instructions for Finance Officer, Alumni and Students

- 1. Finance Officer: open Donations, select the alumnus and campaign, enter the amount, date and payment mode and submit. Amounts below Rs. 100 and future dates are rejected.
- 2. Alumni and students: register for events, post jobs or submit mentorship requests from the corresponding forms and follow their status in the reports.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	C YASHWANTH	Register No.	241501251
Project Title	Alumni Management System	Date	_____

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder pages	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	C YASHWANTH	Register No.	241501251
Department / Section	Artificial Intelligence and Machine Learning	Date of Viva	To be completed

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER EXTERNAL EXAMINER